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MULTI-PIECE SOLID GOLF BALL

Abstract

A multi-piece solid golf ball includes a core, an intermediate layer, and a cover, in which a large number of dimples are formed on an outside surface of the cover, a relationship between a surface hardness of an intermediate layer-encased sphere and a surface hardness of the ball is specified, an initial velocity of the ball is set from 76.0 to 77.724 m/s, a relationship of a ratio to a lift coefficient/drag coefficient at a Reynolds number of 218,000 and a spin rate of 2,800 rpm, a lift coefficient/drag coefficient at a Reynolds number of 184,000 and a spin rate of 2,900 rpm, and a lift coefficient/drag coefficient at a Reynolds number of 158,000 and a spin rate of 3,100 rpm is specified, and a volume occupancy ratio of the dimples is set within a predetermined range.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119 (a) on Patent Application No. 2024-033917 filed in Japan on Mar. 6, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to a multi-piece solid golf ball which, while seeking to comply with new Overall Distance Standard (ODS) rules, which will apply from January 2028 onward, has good flight and rises easily on approach shots by amateur users, and which pursues ease of play in the game of golf.

BACKGROUND ART

[0003] In March 2022, manufacturers of golf balls were notified by the Royal and Ancient Golf Club of St Andrews (hereinafter, R&A) and the United States Golf Association (hereinafter, USGA) that the R&A and the USGA would start research to suppress a distance by the longest hitters by changing test conditions for the Overall Distance Standard (hereinafter, ODS) of golf balls in the future. Further, the R&A and the USGA specifically announced the following in December 2023.

[0004] “Club head speeds in ODS for rule compliance will be increased from the current 54 m/s to 56 m/s. That is, an influence of 13 to 15 yards (11.9 to 13.7 m) is predicted under striking conditions of the longest hitters.”

[0005] This change to the test is to be implemented from January 2028, but it is said that the current test conditions may be used until Jan. 1, 2030 for recreational golf, which is non-competitive.

[0006] Therefore, under the striking conditions of the longest hitters having a high head speed, the time is approaching when the ball as described above with a suppressed distance compared to current balls is required. On the other hand, as a golf ball for amateur users, it is desired that even a ball in which the above-described distance is suppressed has as long a distance as possible under striking conditions at a head speed of amateur users. In addition, since it is difficult for an amateur user who is non-competitive to make the ball rise on approach shots, making it easy for the ball to rise on approach shots, that is, raising a launch angle, makes it easy for the ball to be regarded as “a ball that makes approach shots easy”.

[0007] In addition, in the past, some golf balls in which an initial velocity of the ball is set lower than that of a normal game ball have been proposed. Examples of such technical documents include the following Patent Documents 1 to 5.

[0008] However, each of the proposed golf balls is a practice ball for a driving range that is simply designed so as not to have a larger distance than a game ball. Therefore, in each of the above-mentioned Patent Documents, good distance performance on shots by amateur users and ease of making the ball rise on approach shots are not considered at all.

[0009] Further, Patent Documents 6 to 14 listed below each disclose a golf ball in which, as for dimples formed on the ball surface, a sum of the volumes of the individual dimples, formed below the flat plane circumscribed by the edge of a dimple, to a ball spherical volume on the assumption that the ball has no dimples, that is, a dimple volume occupancy ratio VR, is specified within a predetermined range, whereby a superior distance may be obtained in the low head speed (HS) range while reducing the distance in the high HS range.

[0010] However, with respect to the proposed golf balls, the reducing of the distance in the high

head speed range is too great, and a distance on full shots by amateur users is not satisfactory. Further, in each of the above Patent Documents, ease of making the ball rise on approach shots is not considered at all.

CITATION LIST

[0011] Patent Document 1: JP-A 2012-228470 [0012] Patent Document 2: JP-A 2014-069045
[0013] Patent Document 3: JP-A 2013-138857 [0014] Patent Document 4: JP-A 2013-138839
[0015] Patent Document 5: JP-A 2013-138840 [0016] Patent Document 6: JP-A 2011-218160
[0017] Patent Document 7: JP-A 2011-218161 [0018] Patent Document 8: JP-A 2011-218162
[0019] Patent Document 9: JP-A 2011-240122 [0020] Patent Document 10: JP-A 2011-240123
[0021] Patent Document 11: JP-A 2011-240124 [0022] Patent Document 12: JP-A 2011-240125
[0023] Patent Document 13: JP-A 2011-240126 [0024] Patent Document 14: JP-A 2011-240127

SUMMARY OF THE INVENTION

[0025] The present invention has been made in view of the above circumstances, and an object of the present invention is to provide a multi-piece solid golf ball for conforming to new ODS rules to be applied from January 2028, in which suppression of a distance on striking by the longest hitters is not only caused to at least a specific level, but also distances on full shots with a driver (W #1) and an iron by amateur users may be favorably maintained without decreasing, and moreover, a launch angle on approach shots by amateur users becomes high, and the ball may be made to rise easily.

[0026] As a result of intensive studies to achieve the above object, the present inventor has found that in a multi-piece solid golf ball including a core, an intermediate layer, and a cover, in which a large number of dimples are formed on an outside surface of the cover, a relationship between a surface hardness of an intermediate layer-encased sphere and a surface hardness of the ball satisfies the following condition:

Surface hardness of ball > surface hardness of intermediate layer-encased sphere [0027] where hardness means Shore C hardness, [0028] an initial velocity of the ball is set to from 76.0 to 77.724 m/s, and where a ratio CL1/CD1 of a lift coefficient CL1 at a Reynolds number of 218,000 and a spin rate of 2,800 rpm to a drag coefficient CD1 is denoted by A1, a ratio CL2/CD2 of a lift coefficient CL2 at a Reynolds number of 184,000 and a spin rate of 2,900 rpm to a drag coefficient CD2 is denoted by A2, and a ratio CL3/CD3 of a lift coefficient CL3 at a Reynolds number of 158,000 and a spin rate of 3,100 rpm to a drag coefficient CD3 is denoted by A3, the following two conditions are satisfied:

[00001] $0.59 \leq A1 \leq 0.655$, and $(A2 + A3) / 2 \geq 0.670$.

[0029] Further, the present inventor has found that by designing a golf ball so that a volume occupancy ratio VR of the dimples is from 0.75 to 0.89%, and the following condition is satisfied:

[00002] $100 \leq D / B \leq 140$

[0030] where a total volume of the dimples is denoted by D (mm.sup.3) and a deflection when the ball is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is denoted by B (mm), it is possible to conform to the new ODS rules, that is, a distance on striking by the longest hitters is suppressed to at least a specific level, distances on full shots with a driver (W #1) and an iron by amateur users do not decrease, and conversely, the distances increase. In addition, the present inventor has found that in the above-described designed multi-piece solid golf ball, a launch angle of the ball is increased on approach shots by amateur users, the ball rises easily, the ball becomes a ball that is “easy on approach shots”, while such design that can easily obtain a superior distance on full shots by amateur users is available, and has completed the present invention.

[0031] The above “longest hitters” refer to golfers whose head speed on shots with a driver (W #1) is at least 50 m/s, and the above “non-competitive amateur golfers” refer to golfers who have a head speed with a driver (W #1) of not more than 45 m/s and a handicap of approximately 30 or

more.

[0032] Accordingly, the present invention provides a multi-piece solid golf ball including [0033] a core, an intermediate layer, and a cover, in which a large number of dimples are formed on an outside surface of the cover, a relationship between a surface hardness of an intermediate layer-encased sphere and a surface hardness of the ball satisfies the following condition:

Surface hardness of ball > surface hardness of intermediate layer-encased sphere [0034] where hardness means Shore C hardness, [0035] an initial velocity of the ball is from 76.0 to 77.724 m/s, and where a ratio CL1/CD1 of a lift coefficient CL1 at a Reynolds number of 218,000 and a spin rate of 2,800 rpm to a drag coefficient CD1 is denoted by A1, a ratio CL2/CD2 of a lift coefficient CL2 at a Reynolds number of 184,000 and a spin rate of 2,900 rpm to a drag coefficient CD2 is denoted by A2, and a ratio CL3/CD3 of a lift coefficient CL3 at a Reynolds number of 158,000 and a spin rate of 3,100 rpm to a drag coefficient CD3 is denoted by A3, the following two conditions are satisfied:

[00003] $0.59 \leq A1 \leq 0.655$, and $(A2 + A3) / 2 \geq 0.670$,

and [0036] a volume occupancy ratio VR of the dimples is from 0.75 to 0.89%, and the following condition is satisfied:

[00004] $100 \leq D / B \leq 140$ [0037] where a total volume of the dimples is denoted by D (mm.sup.3) and a deflection when the ball is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is denoted by B (mm).

[0038] In a preferred embodiment of the multi-piece solid golf ball according to the invention, the value of A1 is from 0.590 to 0.613, the value of A2 is from 0.635 to 0.668, and the value of A3 is from 0.695 to 0.734.

[0039] In another preferred embodiment, the value of A1 is from 0.614 to 0.655, the value of A2 is from 0.669 to 0.750, and the value of A3 is from 0.735 to 0.815.

[0040] In still another preferred embodiment, the value of $(A2 + A3) / 2$ is from 0.670 to 0.783.

[0041] In a yet further preferred embodiment, the cover is formed of an ionomer resin as a chief material.

[0042] In a yet further preferred embodiment, the core has a hardness profile in which, letting the Shore C hardness at a core center be Cc, the Shore C hardness at a midpoint M between the core center and a core surface be Cm, the Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm inward from the midpoint M be Cm-2, Cm-4, and Cm-6 respectively, the Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm outward from the midpoint M be Cm+2, Cm+4, and Cm+6 respectively, and the Shore C hardness at the core surface be Cs, and defining surface areas A to F as follows:

$\frac{1}{2} \times 2 \times (Cm - 4 - Cm - 6)$ surface area A:

$\frac{1}{2} \times 2 \times (Cm - 2 - Cm - 4)$ surface area B:

$\frac{1}{2} \times 2 \times (Cm - Cm - 2)$ surface area C:

$\frac{1}{2} \times 2 \times (Cm + 2 - Cm)$ surface area D:

$\frac{1}{2} \times 2 \times (Cm + 4 - Cm + 2)$ surface area E:

$\frac{1}{2} \times 2 \times (Cm + 6 - Cm + 4)$ surface area F: [0043] the following condition is satisfied:

$\{(\text{surface area D} + \text{surface area E}) - (\text{surface area A} + \text{surface area B})\} \geq 4.0$.

[0044] In another preferred embodiment, the core has a hardness profile in which the following

condition is satisfied:

[00005](Cs - Cc) ≥ 20 .

[0045] In a still further preferred embodiment, the core has a hardness profile in which the following condition is satisfied:

[00006](Cs - Cc) / (Cm - Cc) ≥ 3 . 0 .

[0046] In a yet further preferred embodiment, in the core hardness profile, an upper limit of the formula {(surface area D+surface area E)–(surface area A+surface area B)} is 15.0.

[0047] In yet another preferred embodiment, the core is formed of a rubber composition containing the following components (A) to (D): [0048] (A) a base rubber, [0049] (B) an organic peroxide, [0050] (C) water or a monocarboxylic acid metal salt, and [0051] (D) sulfur.

Advantageous Effects of the Invention

[0052] The multi-piece solid golf ball of the present invention is a golf ball that is intended to conform to the new ODS rules to be applied from January 2028, and may increase distances on full shots with a driver (W #1) and an iron by amateur users while reducing the distance on shots with a driver by the longest hitters. In addition, in the golf ball of the present invention, the launch angle of the ball becomes high on approach shots by amateur users, the ball is made to rise easily, and easiness can be felt on approach shots.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0053] FIG. 1 is a schematic cross-sectional view of a golf ball according to one embodiment of the present invention.

[0054] FIG. 2 is a graph that uses core hardness profile data in Example 1 to describe surface areas A to F in the core hardness profile.

[0055] FIG. 3 is a graph showing core hardness profiles in Examples 1 to 4 and Comparative Examples 1 to 4.

[0056] FIG. 4 is a graph showing core hardness profiles in Comparative Examples 5 to 13.

[0057] FIGS. 5A and 5B show an arrangement mode (pattern) of dimples (1) to (5) used in Examples 1 to 4 and Comparative Examples 1 to 13, where FIG. 5A shows a plan view of the dimples, and FIG. 5B shows a side view thereof.

DETAILED DESCRIPTION OF THE INVENTION

[0058] Hereinafter, the present invention is described in more detail.

[0059] A multi-piece solid golf ball according to the present invention has a core, an intermediate layer, and a cover, and an example thereof is shown in FIG. 1. A golf ball G shown in FIG. 1 has a single-layer core 1, a single-layer intermediate layer 2 encasing the core 1, and a single-layer cover 3 encasing the intermediate layer. The cover 3 is positioned at the outermost layer in the layer construction of the golf ball except for the coating layer. In addition to the single layer as shown in FIG. 1, each layer of the core and the intermediate layer may be formed as a plurality of layers. A large number of dimples D are formed on a surface of the cover 3 (outermost layer) in order to obtain aerodynamic properties as intended properties of the present invention. In addition, although not particularly illustrated, the coating layer is typically formed on the surface of the cover 3. Hereinafter, each of the above layers is described in detail.

[0060] The core is obtained by vulcanizing a rubber composition containing a rubber material as a chief material. If the core material is not the rubber composition, the rebound of the core may become low, and a desired distance on shots with a driver (W #1) and an iron by amateur users may not be attainable. The rubber composition typically contains a base rubber as the chief material, and is obtained with the inclusion of a co-crosslinking agent, a co-crosslinking initiator, an inert filler, an organosulfur compound, or the like.

[0061] In particular, the core is suitably formed of a rubber composition containing the following components (A) to (D): [0062] (A) a base rubber, [0063] (B) an organic peroxide, [0064] (C) water or a monocarboxylic acid metal salt, and [0065] (D) sulfur.

[0066] The base rubber (A) may include a diene rubber. Examples of the diene rubber include polybutadiene, natural rubber, isoprene rubber, and ethylene propylene diene rubber.

[0067] As the organic peroxide (B), an organic peroxide having a relatively high thermal decomposition temperature is suitably used. Specifically, a high-temperature organic peroxide having a one-minute half-life temperature of about 165 to 185° C. is used, and examples thereof include dialkyl peroxides. Examples of the dialkyl peroxides include dicumyl peroxide ("Percumyl D" manufactured by NOF Corporation), 2,5-dimethyl-2,5-di(t-butylperoxy) hexane ("Perhexa 25B" manufactured by NOF Corporation), and di(2-t-butylperoxyisopropyl)benzene ("Perbutyl P" manufactured by NOF Corporation), and dicumyl peroxide may be suitably used. These may be used singly, or two or more may be used in combination. The half-life is one of the indices representing a degree of a decomposition rate of the organic peroxide, and is indicated by a time required for the original organic peroxide to be decomposed and its active oxygen amount to reach ½. A vulcanization temperature in the core-forming rubber composition is typically within a range of 120 to 190° C., and in that range, an organic peroxide having a one-minute half-life temperature of a high temperature, which is about 165° C. to 185° C., is thermally decomposed relatively slowly. With the rubber composition used in the present invention, by adjusting the amount of free radicals produced, which increases with the lapse of a vulcanization time, it is possible to obtain a core that is a rubber cross-linked product having a specific internal hardness shape described later.

[0068] The water (C), although not particularly limited, may be distilled water or tap water, but it is particularly suitable to employ distilled water free of impurities. The compounding amount of the water included per 100 parts by weight of the base rubber is preferably at least 0.1 part by weight, and more preferably at least 0.2 parts by weight, and an upper limit thereof is preferably not more than 2 parts by weight, and more preferably not more than 1 part by weight.

[0069] By blending the water or a material containing water as the component (C) directly into the core material, a decomposition of the organic peroxide during the core formulation may be promoted. In addition, it is known that the decomposition efficiency of the organic peroxide in the core-forming rubber composition changes depending on temperature, and the decomposition efficiency increases as the temperature becomes higher than a certain temperature. If the temperature is too high, the amount of decomposed radicals becomes too large, and the radicals are recombined or deactivated. As a result, fewer radicals act effectively in crosslinking. Here, when decomposition heat is generated by the decomposition of the organic peroxide at the time of core vulcanization, a temperature near the core surface is maintained at substantially the same level as a temperature of a vulcanization mold, but the temperature around the core center is considerably higher than the mold temperature due to an accumulation of decomposition heat by the organic peroxide decomposing from the outside. If the water or a material containing water is directly included in the core, the water acts to promote the decomposition of the organic peroxide, so that the radical reactions as described above can be changed at the core center and the core surface.

That is, the decomposition of the organic peroxide is further promoted near the core center, and the deactivation of radicals is further promoted, so that the amount of active radicals is further reduced, and as a result, a core may be obtained in which the crosslink densities at the core center and the core surface differ markedly, and the dynamic viscoelasticity of the core center portion is different.

[0070] In addition, a monocarboxylic acid metal salt may be employed instead of the water. In the monocarboxylic acid metal salt, it is presumed that a carboxylic acid is coordinate-bonded to the metal salt, and the monocarboxylic acid metal salt is distinguished from a dicarboxylic acid metal salt such as zinc diacrylate, which is represented by chemical formula

[CH.sub.2=CHCOO].sub.2Zn. The monocarboxylic acid metal salt brings water into the rubber composition by a dehydration condensation reaction, so that the same effect as that of the water

may be obtained. In addition, since the monocarboxylic acid metal salt may be blended into the rubber composition as powder, the working process may be simplified, and it is easy to uniformly disperse the monocarboxylic acid metal salt in the rubber composition. In order to effectively perform the above reaction, it is necessary to use a mono-salt. The compounding amount of the monocarboxylic acid metal salt is preferably at least 1 part by weight, and more preferably at least 3 parts by weight per 100 parts by weight of the base rubber. As an upper limit thereof, the compounding amount of the monocarboxylic acid metal salt is preferably not more than 60 parts by weight, and more preferably not more than 50 parts by weight per 100 parts by weight of the base rubber. If the compounding amount of the monocarboxylic acid metal salt is too small, it is difficult to obtain an appropriate crosslinking density, and it may not be possible to obtain an adequate golf ball spin rate-lowering effect. In addition, if the compounding amount is too large, the core becomes too hard, so that it may be difficult to maintain an appropriate feel at impact.

[0071] As the carboxylic acid, acrylic acid, methacrylic acid, maleic acid, fumaric acid, stearic acid, or the like may be used. Examples of substitute metal include Na, K, Li, Zn, Cu, Mg, Ca, Co, Ni, and Pb, and Zn is preferably used. Specific examples thereof include zinc monoacrylate and zinc monomethacrylate, and it is particularly preferable to use a zinc monoacrylate.

[0072] Specific examples of the sulfur (D) include trade names "SANMIX S-80N" (manufactured by Sanshin Chemical Industry Co., Ltd.) and "SULFAX-5" (manufactured by Tsurumi Chemical Industry Co., Ltd.). The compounding amount of the sulfur may exceed 0, and may be preferably at least 0.005 parts by weight, and even more preferably at least 0.01 parts by weight per 100 parts by weight of the base rubber. In addition, an upper limit of the compounding amount is not particularly limited, but the upper limit is preferably not more than 0.1 parts by weight, more preferably not more than 0.05 parts by weight, and even more preferably not more than 0.03 parts by weight. The addition of the sulfur may increase a difference in hardness of the core. If the compounding amount of the sulfur is too large, rebound may be greatly reduced, or a durability on repeated impact may worsen.

[0073] In the rubber composition, a co-crosslinking agent, a filler, an antioxidant, an organosulfur compound, and the like may be included as components other than the components (A) to (D).

[0074] The co-crosslinking agent is an α,β -unsaturated carboxylic acid and/or a metal salt thereof. Specific examples of the unsaturated carboxylic acid include acrylic acid, methacrylic acid, maleic acid, fumaric acid, or the like, and in particular, acrylic acid and methacrylic acid are suitably used. The metal salt of the unsaturated carboxylic acid is not particularly limited, and examples thereof include those obtained by neutralizing the unsaturated carboxylic acid with a desired metal ion. Specific examples thereof include zinc salts and magnesium salts such as methacrylic acid and acrylic acid, and in particular, zinc acrylate is suitably used.

[0075] The unsaturated carboxylic acid and/or the metal salt thereof is typically blended in an amount of at least 5 parts by weight, preferably at least 9 parts by weight, and even more preferably at least 13 parts by weight, and the upper limit is typically not more than 60 parts by weight, preferably not more than 50 parts by weight, and even more preferably not more than 40 parts by weight per 100 parts by weight of the base rubber. If the compounding amount is too large, the core may become too hard, giving the ball an unpleasant feel at impact, and if the compounding amount is too small, rebound may become low.

[0076] As a filler, for example, zinc oxide, barium sulfate, calcium carbonate, or the like may be suitably used. These may be used singly, or two or more may be used in combination. The compounding amount of the filler may be preferably at least 4 parts by weight, more preferably at least 8 parts by weight, and even more preferably at least 11 parts by weight per 100 parts by weight of the base rubber. In addition, an upper limit of the compounding amount is preferably not more than 50 parts by weight, more preferably not more than 40 parts by weight, and even more preferably not more than 30 parts by weight per 100 parts by weight of the base rubber. If the compounding amount is too large or too small, it may not be possible to obtain an appropriate

weight and a suitable rebound.

[0077] As an antioxidant, for example, commercially available products such as Nocrac NS-6, Nocrac NS-30, Nocrac NS-200, and Nocrac MB (all manufactured by Ouchi Shinko Chemical Industrial Co., Ltd.) may be employed. These may be used singly, or two or more may be used in combination.

[0078] The compounding amount of the antioxidant is not particularly limited, but is preferably 0.05 parts by weight or more, and more preferably 0.1 parts by weight or more, and the upper limit is preferably 1.0 part by weight or less, more preferably 0.7 parts by weight or less, and even more preferably 0.5 parts by weight or less per 100 parts by weight of the base rubber. If the compounding amount is too large or too small, a suitable core hardness gradient cannot be obtained, and it may not be possible to obtain suitable rebound, durability, and a spin rate-lowering effect on full shots.

[0079] The organosulfur compound may be included in order to control the rebound of the core so that it is increased. As the organosulfur compound, specifically, it is recommended to include thiophenol, thionaphthol, halogenated thiophenol, or the metal salt thereof. More specifically, the examples of the organosulfur compound include zinc salts such as pentachlorothiophenol, pentafluorothiophenol, pentabromothiophenol, p-chlorothiophenol, and pentachlorothiophenol, and any of the following having 2 to 4 sulfur atoms: diphenylpolysulfide, dibenzylpolysulfide, dibenzoylpolysulfide, dibenzothiazoylpolysulfide, and dithiobenzoylpolysulfide. In particular, diphenyldisulfide and the zinc salt of pentachlorothiophenol is preferably used.

[0080] An upper limit of a compounding amount of the organosulfur compound is preferably not more than 5 parts by weight, more preferably not more than 4 parts by weight, even more preferably not more than 3 parts by weight, and most preferably not more than 2 parts by weight per 100 parts by weight of the base rubber. If the compounding amount is too large, the core hardness becomes too soft or the rebound of the core becomes too high, and a distance on shots with a driver by the longest hitters may be too long.

[0081] The core can be manufactured by vulcanizing and curing the rubber composition containing the above components. For example, a molded body can be manufactured by intensively mixing the rubber composition using a mixing apparatus such as a Banbury mixer or a roll mill, subsequently compression molding or injection molding the mixture using a core mold, and curing the resulting molded body by appropriately heating it at a temperature sufficient for the organic peroxide or the co-crosslinking agent to act, such as at a temperature of 100 to 200° C., and preferably at a temperature of 140 to 180° C., for 10 to 40 minutes.

[0082] In the present invention, the core is formed as a single layer or a plurality of layers, although it is preferably formed as a single layer. If a rubber core is produced as a plurality of layers of rubber, layer separation at an interface may arise when the ball is repeatedly struck, possibly leading to cracking at an earlier stage.

[0083] The diameter of the core is preferably at least 35.1 mm, more preferably at least 36.7 mm, and even more preferably at least 37.3 mm. The upper limit of the diameter of the core is preferably not more than 38.7 mm, more preferably not more than 38.3 mm, and even more preferably not more than 37.9 mm. If the core diameter is too small, an initial velocity of the ball may become too low, or a deflection of an entire ball may become small, a spin rate of the ball on full shots may increase, and the desired distance on full shots by amateur users may not be attainable. On the other hand, if the core diameter is too large, the spin rate on full shots increases, and the desired distance may not be attainable by amateur users, or a durability to cracking on repeated impact may worsen.

[0084] The deflection (mm) when the core is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is not particularly limited, but is preferably at least 2.8 mm, more preferably at least 3.0 mm, and even more preferably at least 3.2 mm, and the upper limit thereof is preferably not more than 4.5 mm, more preferably not more than 4.0 mm, and even more preferably not more than 3.5 mm. If the deflection of the core is too small, that is, if the core is too

hard, the spin rate on full shots increases excessively, the distance on shots with a driver (W #1) and an iron by amateur users may not be sufficiently increased, and the feel at impact may be too hard. On the other hand, if the deflection of the core is too large, that is, if the core is too soft, an actual initial velocity becomes too low, so that the distance on shots with a driver (W #1) by the longest hitters and amateur users may be too short, the feel at impact may become too soft, and the durability to cracking on repeated impact may become too poor.

[0085] Next, the core hardness profile is described. It is noted that the hardness of the core described below means Shore C hardness. The Shore C hardness is a hardness value measured with a Shore C durometer conforming to the ASTM D2240 standard.

[0086] A core center hardness (Cc) is preferably at least 57, more preferably at least 59, and even more preferably at least 61, and the upper limit is preferably not more than 67, more preferably not more than 65, and even more preferably not more than 63. If this value is too large, the spin rate of the ball on full shots may rise, the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the feel at impact may be too hard. On the other hand, if the above value is too small, the rebound may become low, and the desired distance for amateur users may not be attainable, or the durability to cracking on repeated impact may worsen.

[0087] A hardness (Cm-6) at a position 6 mm inward from the point M (hereinafter, also referred to as "midpoint M") between the core center and the core surface is not particularly limited, but may be preferably at least 57, more preferably at least 59, and even more preferably at least 61, and the upper limit is also not particularly limited, and may be preferably not more than 67, more preferably not more than 65, and even more preferably not more than 63. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core center hardness (Cc).

[0088] A hardness (Cm-4) at a position 4 mm inward from the point M between the core center and the core surface is not particularly limited, but may be preferably at least 58, more preferably at least 60, and even more preferably at least 62, and the upper limit is also not particularly limited, and may be preferably not more than 66, more preferably not more than 68, and even more preferably not more than 64. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core center hardness (Cc).

[0089] A hardness (Cm-2) at a position 2 mm inward from the midpoint M of the core is not particularly limited, but may be preferably at least 58, more preferably at least 60, and even more preferably at least 62. The upper limit is also not particularly limited, and may be preferably not more than 69, more preferably not more than 67, and even more preferably not more than 65. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core center hardness (Cc).

[0090] A cross-sectional hardness (Cm) at the midpoint M of the core is not particularly limited, but may be preferably at least 59, more preferably at least 61, and even more preferably at least 63. In addition, although not particularly limited, an upper limit thereof may be preferably not more than 69, more preferably not more than 67, and even more preferably not more than 65. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core center hardness (Cc).

[0091] A core surface hardness (Cs) is preferably at least 81, more preferably at least 83, and even more preferably at least 85. The upper limit value is preferably not more than 91, more preferably not more than 89, and even more preferably not more than 87. If this value is too large, the durability to cracking on repeated impact may worsen, or the feel at impact may be too hard. On the other hand, if the above value is too small, the rebound may become low or the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users.

[0092] A hardness (Cm+2) at a position 2 mm outward toward the core surface (hereinafter, simply referred to as "outward") from the midpoint M of the core toward the core surface is not

particularly limited, but may be preferably at least 60, more preferably at least 62, and even more preferably at least 64. The upper limit is also not particularly limited, and may be preferably not more than 70, more preferably not more than 68, and even more preferably not more than 66. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core surface hardness (Cs).

[0093] A hardness (Cm+4) at a position 4 mm outward from the midpoint M of the core is not particularly limited, but may be preferably at least 65, more preferably at least 67, and even more preferably at least 69. The upper limit is also not particularly limited, and may be preferably not more than 75, more preferably not more than 73, and even more preferably not more than 71. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core surface hardness (Cs).

[0094] A hardness (Cm+6) at a position 6 mm outward from the midpoint M of the core is not particularly limited, but may be preferably at least 71, more preferably at least 73, and even more preferably at least 75. The upper limit is also not particularly limited, and may be preferably not more than 81, more preferably not more than 79, and even more preferably not more than 77. Hardnesses that deviate from these values may lead to undesirable results similar to those described above for the core surface hardness (Cs).

[0095] A value obtained by subtracting the core center hardness from the core surface hardness, that is, the value of Cs-Cc, is preferably at least 20, more preferably at least 21, and even more preferably at least 22, and an upper limit thereof is preferably not more than 30, more preferably not more than 27, and even more preferably not more than 24. If this value is too small, the spin rate of the ball on full shots may rise, the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the feel at impact may be too hard. On the other hand, if this value is too large, the rebound may become low, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the durability to cracking on repeated impact may worsen.

[0096] In addition, it is preferable to optimize the value of (Cs-Cc)/(Cm-Cc) for the core hardness profile. The value of (Cs-Cc) indicates a difference in hardness between the core center and the core surface, and the value of (Cm-Cc) indicates a difference in hardness between the core center and the midpoint between the core surface and the core center, and the above expression represents the ratio of these differences in hardness. The value of (Cs-Cc)/(Cm-Cc) is preferably at least 3.0, more preferably at least 5.0, and even more preferably at least 7.0. If this value is too small, the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users.

[0097] In the core hardness profile, surface areas A to F are defined as follows:

$\frac{1}{2} \times 2 \times (Cm-4 - Cm-6)$ surface area A:

$\frac{1}{2} \times 2 \times (Cm-2 - Cm-4)$ surface area B:

$\frac{1}{2} \times 2 \times (Cm - Cm-2)$ surface area C:

$\frac{1}{2} \times 2 \times (Cm+2 - Cm)$ surface area D:

$\frac{1}{2} \times 2 \times (Cm+4 - Cm+2)$ surface area E:

$\frac{1}{2} \times 2 \times (Cm+6 - Cm+4)$ surface area F: [0098] and are characterized in that a value of (surface area D+surface area E)-(surface area A+surface area B) is preferably at least 4.0, more preferably at least 4.5, and even more preferably at least 5.0, and the upper limit is preferably not more than 15.0, more preferably not more than 10.0, and even more preferably not more than 7.0. If this value

is too small, the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users. On the other hand, if this value is too large, the rebound may become low, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the durability to cracking on repeated impact may worsen.

[0099] A value of (surface area D+surface area E)–(surface area B+surface area C) is preferably at least 4.0, more preferably at least 4.5, and even more preferably at least 5.0, and the upper limit is preferably not more than 15.0, more preferably not more than 10.0, and even more preferably not more than 7.0. If this value is too small, the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users. On the other hand, if this value is too large, the rebound may become low, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the durability to cracking on repeated impact may worsen.

[0100] A value of {(surface area D+surface area E)–(surface area A+surface area B)}×(Cs–Cc) is preferably at least 80, more preferably at least 100, and even more preferably at least 120, and the upper limit is preferably not more than 300, more preferably not more than 180, and even more preferably not more than 140. In addition, a value of {(surface area D+surface area E)–(surface area B+surface area C)}×(Cs–Cc) is preferably at least 80, more preferably at least 100, and even more preferably at least 120, and the upper limit is preferably not more than 300, more preferably not more than 180, and even more preferably not more than 140. In addition, if these values are too small, the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users. On the other hand, if these values are too large, the rebound may become low, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users, or the durability to cracking on repeated impact may worsen.

[0101] A relationship between the surface areas calculated from the hardness profile described above is preferably surface area F>surface area D>surface area A, more preferably surface area F>surface area E>surface area D>surface area A, and even more preferably surface area F>surface area E>surface area D>(surface area A+surface area B). If these relational expressions are not satisfied, the spin rate of the ball on full shots may rise, and the desired distance may not be attainable on shots with a driver (W #1) and an iron by amateur users.

[0102] FIG. 2 shows a graph describing the surface areas A to F using the core hardness profile data of Example 1. In this way, the surface areas A to F are surface areas of each triangle whose base is a difference between each specific distance and whose height is a difference in hardness between each position at these specific distances.

[0103] Next, the intermediate layer is described.

[0104] The intermediate layer has a material hardness on the Shore C hardness scale which, although not particularly limited, is preferably at least 71, more preferably at least 78 and even more preferably at least 84, and the upper limit is preferably not more than 95, more preferably not more than 92 and even more preferably not more than 90. The surface hardness on the Shore D hardness scale is preferably at least 46, more preferably at least 50 and even more preferably at least 53, and the upper limit is preferably not more than 61, more preferably not more than 60 and even more preferably not more than 59.

[0105] The intermediate layer-encased sphere obtained by encasing the core with the intermediate layer has a surface hardness which, on the Shore C hardness scale, is preferably at least 80, more preferably at least 85, and even more preferably at least 90. The upper limit is preferably not more than 96, more preferably not more than 95, and even more preferably not more than 94. The surface hardness on the Shore D hardness scale is preferably at least 52, more preferably at least 56 and even more preferably at least 59, and the upper limit is preferably not more than 67, more preferably not more than 65 and even more preferably not more than 63.

[0106] If the material hardness and the surface hardness of the intermediate layer are too soft in comparison with the above ranges, the spin rate of the ball on full shots may rise, and the distance on shots with a driver (W #1) and an iron by amateur users may not be increased. In addition, the launch angle on approach shots may become low, and the ball may be felt by amateur users to be difficult to handle. On the other hand, if the material hardness and the surface hardness of the intermediate layer are too hard in comparison with the above ranges, the feel at impact may become too hard, the durability to cracking on repeated impact may worsen, and the distance on shots by the longest hitters may become too long.

[0107] The intermediate layer has a thickness which is preferably at least 0.8 mm, more preferably at least 1.0 mm, and even more preferably at least 1.2 mm. The intermediate layer thickness has an upper limit that is preferably not more than 1.5 mm, more preferably not more than 1.4 mm, and even more preferably not more than 1.35 mm. If the intermediate layer is too thin, the durability to cracking on repeated impact may worsen, or it may be difficult to mold the cover and mass productivity may be deteriorated. On the other hand, if the intermediate layer is too thick, the feel at impact may become too hard, the rebound may become low, and it may be difficult for the distance on shots by amateur users to increase. In addition, if the cover thickness deviates from the above ranges, the spin rate of the ball on full shots may rise, and a good distance may not be increased on shots by amateur users.

[0108] As a material of the intermediate layer, a resin material containing an ionomer resin as a chief material is used, and particularly among ionomer resins, a highly neutralized ionomer is suitably used. As the highly neutralized ionomer, a commercially available product may be used, and examples thereof include “HPF 1000”, “HPF 2000”, “HPF AD1035”, and “HPF AD1040” (all manufactured by the Dow Chemical Company). By adopting the highly neutralized ionomer, the rebound of the ball is increased while reducing the spin rate of the ball on full shots, and a good distance may be reliably obtained on full shots by amateur users.

[0109] Next, the cover is described.

[0110] The cover has a material hardness on the Shore C hardness scale which, although not particularly limited, is preferably at least 83, more preferably at least 89, and even more preferably at least 93, and the upper limit is preferably not more than 100, more preferably not more than 97, and even more preferably not more than 95. The material hardness on the Shore D hardness scale is preferably at least 55, more preferably at least 60, and even more preferably at least 63, and the upper limit is preferably not more than 75, more preferably not more than 70, and even more preferably not more than 68.

[0111] A surface hardness of the ball (whole sphere) including the core and the cover is preferably at least 90, more preferably at least 93, and even more preferably at least 95, and the upper limit thereof is preferably not more than 100, more preferably not more than 99, and even more preferably not more than 98 on the Shore C hardness scale. The surface hardness on the Shore D hardness scale is preferably at least 62, more preferably at least 67, and even more preferably at least 70, and the upper limit thereof is preferably not more than 76, more preferably not more than 74, and even more preferably not more than 72.

[0112] If the material hardness and the surface hardness of the cover are too soft in comparison with the above ranges, the spin rate of the ball on full shots may rise, and the distance on shots with a driver (W #1) and an iron by amateur users may not be increased. In addition, the launch angle on approach shots may become low, and the ball may be felt by amateur users to be difficult to handle. On the other hand, if the material hardness and the surface hardness of the cover are too hard in comparison with the above ranges, the feel at impact may become too hard, the durability to cracking on repeated impact may worsen, and the distance on shots by the longest hitters may become too long.

[0113] The cover has a thickness of preferably at least 0.8 mm, more preferably at least 1.0 mm, and even more preferably at least 1.2 mm. The upper limit in the cover thickness is preferably not

more than 1.5 mm, more preferably not more than 1.4 mm, and even more preferably not more than 1.35 mm. If the cover is too thin, the durability to cracking on repeated impact may worsen, or it may be difficult to mold the cover and mass productivity may be deteriorated. On the other hand, if the cover is too thick, the feel at impact may become too hard, the rebound may become low, and it may be difficult for the distance on shots by amateur users to increase. In addition, if the cover thickness deviates from the above ranges, the spin rate of the ball on full shots may rise, and a good distance may not be increased on shots by amateur users.

[0114] As the material of the cover, a resin material containing an ionomer resin as the chief material is used. If a urethane material is used as the cover material, since the material is not a hard material, the launch angle on approach shots may become low, and the ball may be felt by amateur users to be difficult to handle. In addition, even if a material having the hardest grade among the types of urethane materials is selected, the rebound may be lower than that of an ionomer material having the same hardness, and the distance on shots by amateur users may not reach a target. In addition, when the cover is formed, from the viewpoint of mass productivity, it is preferable to adopt a method of encasing the core or an intermediate layer-encased sphere with the cover by injection molding.

[0115] In the case where a resin material including an ionomer resin as the chief material is employed, an aspect that uses in admixture a zinc-neutralized ionomer resin and a sodium-neutralized ionomer resin as the chief materials is desirable. The blending ratio in terms of zinc-neutralized ionomer resin/sodium-neutralized ionomer resin (weight ratio) is from 5/95 to 95/5, preferably from 20/80 to 90/10, and more preferably from 30/70 to 70/30. If the zinc-neutralized ionomer and the sodium-neutralized ionomer are not included in this ratio, the rebound may become too low to obtain a desired flight, the durability to cracking on repeated impact at room temperature may worsen, and the durability to cracking at a low temperature (below zero) may worsen.

[0116] Various additives may be appropriately included in the cover material as necessary, for example, a pigment, a dispersant, an antioxidant, a light stabilizer, an ultraviolet absorber, an internal mold lubricant, or the like.

[0117] The manufacture of a multi-piece solid golf ball in which the above-described core, intermediate layer, and cover (outermost layer) are formed as successive layers can be performed by a customary method such as a known injection molding process. For example, an intermediate layer material is injected around the core in an injection mold to obtain an intermediate layer-encased sphere, and finally, a cover material, which is the outermost layer, is injection molded to obtain a multi-piece golf ball. In addition, it is also possible to produce a golf ball by preparing two half-cups pre-molded into hemispherical shapes, enclosing the core and the intermediate layer-encased sphere within the two half cups, and molding the core and the intermediate layer-encased sphere under applied heat and pressure.

[0118] The golf ball has a deflection (mm) when compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) which is preferably at least 2.2 mm, more preferably at least 2.5 mm, and even more preferably at least 2.7 mm. On the other hand, an upper limit of the deflection is preferably not more than 3.8 mm, more preferably not more than 3.4 mm, and even more preferably not more than 3.0 mm. If the deflection of the golf ball is too small, the spin rate of the ball on full shots increases excessively, and the distance on shots with a driver (W #1) and an iron by amateur users may not be increased, or the feel at impact may be too hard. On the other hand, if the deflection is too large, the actual initial velocity becomes too low, so that the distance on shots with a driver (W #1) by the longest hitters and amateur users may be too short, the feel at impact may become too soft, or the durability to cracking on repeated impact may become too poor.

[0119] The initial velocity of the ball is preferably at least 76.0 m/s, more preferably at least 76.5 m/s, and even more preferably at least 77.0 m/s. An upper limit thereof is not more than 77.724

m/s. If this initial velocity value is too high, the official rules of R&A and USGA are not satisfied. On the other hand, if the initial velocity is too low, the actual initial velocity may become low and the desired distance may not be attainable under all striking conditions on full shots. The value of the initial velocity in this case is a numerical value measured by a device for measuring a coefficient of restitution (COR) (Golf Ball Testing Machine) of the same type as the R&A. Specifically, a device for measuring a COR manufactured by Hye Precision USA is used. As a condition, at the time of measurement, an air pressure is changed in four stages and measured, a relational expression between the incident velocity and the COR is constructed, and the initial velocity at an incident velocity of 43.83 m/s is determined from the relational expression. For a measurement environment of the device for measuring a COR, a ball temperature-controlled for at least three hours in a thermostatic bath adjusted to $23.9 \pm 1^\circ \text{C}$. is used, and measurement is performed at a room temperature of $23.9 \pm 2^\circ \text{C}$. In addition, a barrel diameter is selected such that a clearance on one side with respect to an outer diameter of the object being measured is from 0.2 to 2.0 mm.

[0120] A value obtained by dividing the initial velocity of the ball by the deflection of the ball is preferably at least 20, more preferably at least 23, and even more preferably at least 26, and the upper limit is preferably not more than 32, more preferably not more than 30, and even more preferably not more than 29. This value has a meaning of measuring a magnitude of the actual initial velocity of the ball. If this value is too large, the distance on shots by the longest hitters becomes too long, and it may be impossible to conform to the new ODS rules. On the other hand, if this value is too small, the spin rate of the ball on full shots may increase, or the actual initial velocity may become lower, and the desired distance may not be attainable under all striking conditions.

[0121] The total thickness of the cover and the intermediate layer is preferably at least 2.0 mm, more preferably at least 2.2 mm, and even more preferably at least 2.4 mm. An upper limit of the total thickness is preferably not more than 3.8 mm, more preferably not more than 3.0 mm, and even more preferably not more than 2.7 mm. If the total thickness is too small, the durability to cracking on repeated impact may worsen, or it may be difficult to mold the intermediate layer and the cover, and as a result, mass productivity of the ball may be deteriorated. On the other hand, if the total thickness is too large, the feel at impact may become too hard, the rebound may become low, and it may be difficult for the distance on shots by amateur users to increase. In addition, if the cover thickness deviates from the above ranges, the spin rate of the ball on full shots may rise, and a good distance may not be increased on shots by amateur users.

[Relationships Between Surface Hardnesses of Each Sphere]

[0122] In the present invention, from the viewpoint that a relationship between the surface hardness of the intermediate layer-encased sphere and the surface hardness of ball is compatible with a superior distance on shots with a driver (W #1) by amateur users and on full shots with an iron, the durability to cracking on repeated impact and, ease of making the ball rise on approach shots the following condition needs to be satisfied:

(surface hardness of ball) > (surface hardness of intermediate layer-encased sphere).

Expressed on the Shore C hardness scale, a value obtained by subtracting the surface hardness of the intermediate layer-encased sphere from the surface hardness of the ball is preferably larger than 0, more preferably at least 2, and even more preferably at least 4. An upper limit thereof is preferably not more than 25, more preferably not more than 15, and even more preferably not more than 10. If the above-described value is too small, the spin rate of the ball on full shots may rise, the intended distance may not be attainable on full shots with a driver (W #1) and an iron by amateur users, or it may be difficult to make the ball rise on approach shots. If the above-described value is too large, the durability to cracking on repeated impact may worsen, the actual initial velocity may become low, and the distance on all shots with a driver (W #1) may be shorter than

the desired distance.

[0123] A value obtained by subtracting the surface hardness of the core from the surface hardness of the intermediate layer-encased sphere is preferably larger than 0, more preferably at least 2, and even more preferably at least 4, and an upper limit thereof is preferably not more than 25, more preferably not more than 15, and even more preferably not more than 10. Expressed on the Shore C hardness scale, a value obtained by subtracting the center hardness of the core from the surface hardness of the intermediate layer-encased sphere is preferably at least 16, more preferably at least 20, and even more preferably at least 24, and an upper limit value thereof is preferably not more than 44, more preferably not more than 38, and even more preferably not more than 32. If these values are too large, the durability to cracking on repeated impact may worsen, the actual initial velocity may become low, and the distance on all shots with a driver (W #1) may be shorter than the desired distance. On the other hand, if these values are too small, the spin rate of the ball on full shots may rise, and the intended distances on full shots with a driver (W #1) and an iron by amateur users may not be attainable.

[0124] Expressed on the Shore C hardness scale, a value obtained by subtracting the core surface hardness from the ball surface hardness is preferably at least 2, more preferably at least 6, and even more preferably at least 10, and an upper limit thereof is preferably not more than 25, more preferably not more than 20, and even more preferably not more than 15. Expressed on the Shore C hardness scale, a value obtained by subtracting the core center hardness from the ball surface hardness is preferably at least 24, more preferably at least 28, and even more preferably at least 32, and an upper limit thereof is preferably not more than 50, more preferably not more than 45, and even more preferably not more than 40. If these values are too large, the durability to cracking on repeated impact may worsen, the actual initial velocity may become low, and the distance on all shots with a driver (W #1) may be shorter than the desired distance. If this value is too small, the spin rate of the ball on full shots may rise, and a desired distance may not be attainable on shots with a driver (W #1) and on shots with an iron by amateur users.

[Relationship of Deflection Between Core and Ball]

[0125] Letting each deflection (mm) when each sphere of the core and the ball is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) be C (mm) and B (mm) respectively, a value of C-B is preferably at least 0.40 mm, more preferably at least 0.50 mm, and even more preferably at least 0.60 mm, and an upper limit thereof is preferably not more than 0.80 mm, more preferably not more than 0.75 mm, and even more preferably not more than 0.70 mm. In addition, a value of C/B is preferably at least 1.08, more preferably at least 1.10, and even more preferably at least 1.20, and the upper limit thereof is preferably not more than 1.40, more preferably not more than 1.35, and even more preferably not more than 1.30. If these values are too large, the durability to cracking on repeated impact may worsen, the actual initial velocity becomes lower, and the distance on all shots with a driver (W #1) may be shorter than the desired distance. If this value is too small, the spin rate of the ball on full shots may rise, and a desired distance may not be attainable on shots with a driver (W #1) and on shots with an iron by amateur users.

[Core Diameter and Ball Diameter]

[0126] A relationship between the core diameter and the ball diameter, that is, a value of (core diameter)/(ball diameter) is preferably at least 0.822, more preferably at least 0.859, and even more preferably at least 0.874. An upper limit value thereof is preferably not more than 0.906, more preferably not more than 0.897, and even more preferably not more than 0.888. If this value is too small, the initial velocity of the ball becomes low, or the deflection of the entire ball becomes small and the ball becomes hard, the spin rate of the ball on full shots increases, and the desired distance on full shots by amateur users may not be attainable. On the other hand, if the above value is too large, the spin rate of the ball on full shots increases, the desired distance on full shots by amateur users may not be attainable, and the durability to cracking on repeated impact may worsen.

[0127] Numerous dimples may be formed on the outside surface of the cover. Although not

particularly limited, the number of dimples arranged on the surface of the cover is preferably at least 280, preferably at least 300, and more preferably at least 310, and the upper limit thereof can be preferably not more than 450, more preferably not more than 400, and even more preferably not more than 350. If the number of dimples deviates from the above ranges, the distance on shots with a driver (W #1) by amateur users may be shortened.

[0128] As for the shape of the dimples, one type or a combination of two or more types such as a circular shape, various polygonal shapes, a dewdrop shape, and other oval shapes can be appropriately used. For example, if circular dimples are used, the diameter can be about 2.5 mm or more and 6.5 mm or less, and the depth can be 0.08 mm or more and 0.30 mm or less.

[0129] A dimple coverage ratio of the dimples on the spherical surface of the golf ball, specifically, a ratio (surface area coverage ratio, hereinafter, SR value) of a sum of the individual dimple surface areas, each defined by a flat plane circumscribed by an edge of a dimple, to a ball spherical surface area on the assumption that the ball has no dimples is preferable at least 75%, more preferably at least 80%, and even more preferably at least 84%. The upper limit is not more than 90%, more preferably not more than 88%, and even more preferably not more than 86%. If the SR value deviates from the above ranges, the distance on shots with a driver (W #1) by amateur users may be shortened.

[0130] A VR value of a sum of volumes of the individual dimples formed below the flat plane circumscribed by the edge of a dimple to a ball spherical volume on the assumption that the ball has no dimples is at least 0.75%, preferably at least 0.78%, and more preferably at least 0.80%. An upper limit thereof is not more than 0.89%, more preferably not more than 0.88%, and even more preferably not more than 0.86%. If this VR value is larger than the above ranges, the distance on shots with a driver (W #1) by the longest hitters may be too short, or the intended distance on shots with a driver (W #1) by amateur users may not be attainable. In addition, in this case, a ball trajectory may become lower, it becomes difficult to carry, and it may become difficult to go over a valley or a pond. On the other hand, if the above value is too small, the distance on shots with a driver (W #1) by the longest hitters does not decrease, and there is a possibility that the ball flies too much further than an upper limit distance of the new ODS rules.

[0131] A total volume of the dimples means the sum of volumes of the individual dimples formed below the flat plane circumscribed by the edge of the dimple in all the dimples formed on one ball. The total volume of the dimples is not particularly limited, but the total volume is preferably at least 306 mm.sup.3, more preferably at least 318 mm.sup.3, and even more preferably at least 326 mm.sup.3, and the upper limit thereof is preferably not more than 363 mm.sup.3, more preferably not more than 359 mm.sup.3, and even more preferably not more than 351 mm.sup.3. If the value of this total volume of the dimples is larger than the above ranges, the distance on shots with a driver (W #1) by the longest hitters may be too short, or the intended distance on shots with a driver (W #1) by amateur users may not be attainable. In addition, in this case, a ball trajectory may become lower, it becomes difficult to carry, and it may become difficult to go over a valley or a pond. On the other hand, if the above value is too small, the distance on shots with a driver (W #1) by the longest hitters does not decrease, and there is a possibility that the ball flies too much further than an upper limit distance of the new ODS rules.

[0132] A value $V_{sub.0}$ obtained by dividing the spatial volume of the dimples below the flat plane circumscribed by the edge of each dimple by a volume of a cylinder whose base is the flat plane and whose height is a maximum depth of the dimple from the base is preferably at least 0.35, more preferably at least 0.38, and further preferably at least 0.40. The upper limit is not more than 0.80, more preferably not more than 0.70, and even more preferably not more than 0.60. If the $V_{sub.0}$ value deviates from the above ranges, the distance on shots with a driver (W #1) by the longest hitters and amateur users may be shorter than the intended distance.

[0133] In the golf ball of the present invention, when a ratio (CL1/CD1) of a lift coefficient CL1 at a Reynolds number of 218000 and a spin rate of 2800 rpm to a drag coefficient CD1 is denoted by

A1, a ratio (CL2/CD2) of a lift coefficient CL2 at a Reynolds number of 184000 and a spin rate of 2900 rpm to a drag coefficient CD2 is denoted by A2, and a ratio (CL3/CD3) of a lift coefficient CL3 at a Reynolds number of 158000 and a spin rate of 3100 rpm to a drag coefficient CD3 is denoted by A3, the dimples are designed to satisfy the following two conditions:

[00007] $0.59 \leq A1 \leq 0.655$, and $(A2 + A3) / 2 \geq 0.67$.

[0134] In the present specification, the “lift coefficients (CL1, CL2, CL3), drag coefficients (CD1, CD2, CD3)” are measured in accordance with the Indoor Test Range (ITR) defined by the USGA (United States Golf Association). The lift coefficients and the drag coefficients can be adjusted by adjusting the configuration of the dimples of the golf ball (arrangement, diameter, depth, volume, number, shape, and the like). The lift coefficients and the drag coefficients are independent of the internal configuration of the golf ball. The Reynolds number (Re) is a dimensionless number used in the field of hydrodynamics. The Reynolds number (Re) is calculated by the following equation (1).

[00008] $Re = \frac{vL}{\mu}$ (1)

[0135] In Equation (1) above, ρ represents the density of a fluid, v represents the average velocity of an object relative to the flow of the fluid, L represents a characteristic length, and μ represents the viscosity coefficient of the fluid.

[0136] In the present invention, when a ratio CL1/CD1 of a lift coefficient CL1 at a Reynolds number of 218000 and a spin rate of 2800 rpm to a drag coefficient CD1 is defined as A1, a ratio CL2/CD2 of a lift coefficient CL2 at a Reynolds number of 184000 and a spin rate of 2900 rpm to a drag coefficient CD2 is defined as A2, and a ratio CL3/CD3 of a lift coefficient CL3 at a Reynolds number of 158000 and a spin rate of 3100 rpm to a drag coefficient CD3 is defined as A3.

[0137] If a condition of the Reynolds number 218,000 and the spin rate 2,800 rpm under which the lift coefficient CL1 and the drag coefficient CD1 are measured is described, this high-speed condition corresponds to a condition provided by the longest hitters with a driver (W #1), this Reynolds number corresponds to a ball speed when the golf ball is driven out at a head speed (HS) of 54 m/s, and the spin rate 2,800 rpm is an average spin condition of a player with a head speed (HS) of 54 m/s.

[0138] If a condition under which the lift coefficient CL2 and the drag coefficient CD2 are measured is described, that is, a Reynolds number of 184,000 and a spin rate of 2,900 rpm, this middle-speed condition corresponds to a condition on shots with a driver (W #1) by amateur users at a head speed (HS) of 45 m/s, this Reynolds number corresponds to the ball speed when the golf ball is struck at a head speed (HS) of 45 m/s, and the spin rate of 2,900 rpm is the average spin condition of a player with a head speed (HS) of 45 m/s.

[0139] If a condition under which the lift coefficient CL3 and the drag coefficient CD3 are measured is described, that is, a Reynolds number of 158,000 and a spin rate of 3,100 rpm, this low-speed condition corresponds to a condition on shots with a driver (W #1) by amateur users at a head speed (HS) of 40 m/s, this Reynolds number corresponds to a ball speed when the golf ball is struck at a head speed (HS) of 40 m/s, and the spin rate of 3,100 rpm is the average spin condition of a player with a head speed (HS) of 40 m/s.

[0140] The ratio between the lift coefficient CL1 and the drag coefficient CD1, that is, the value of CL1/CD1=A1 is at least 0.590, preferably at least 0.595, and more preferably at least 0.600, and an upper limit thereof is not more than 0.655, preferably not more than 0.640, and more preferably not more than 0.627. If this value is too large, the distance on shots with a driver (W #1) by the longest hitters does not decrease, and there is a possibility that the ball flies too much further than the upper limit distance of the new ODS rules. On the other hand, if the above value is too small, the distance may become too short compared to the intended distance under all striking conditions.

[0141] When the value of A1 is from 0.590 to 0.613, the ratio between the lift coefficient CL2 and

the drag coefficient CD2, that is, a value of $CL2/CD2=A2$, is preferably at least 0.635, more preferably at least 0.645, and even more preferably at least 0.655, and an upper limit thereof is preferably not more than 0.668, more preferably not more than 0.666, and even more preferably not more than 0.664. When the value of A1 is from 0.614 to 0.655, the value of A2 is preferably at least 0.669, more preferably at least 0.671, and even more preferably at least 0.673, and an upper limit thereof is preferably not more than 0.750, more preferably not more than 0.725, and even more preferably not more than 0.700. If the above value deviates from the above ranges, under all striking conditions, the ball may blow up, there may be a trajectory in which the ball does not carry, and an intended total distance may not be attainable.

[0142] When the value of A1 is from 0.590 to 0.613, the ratio between the lift coefficient CL3 and the drag coefficient CD3, that is, a value of $CL3/CD3=A3$, is preferably at least 0.695, more preferably at least 0.705, and even more preferably at least 0.715, and an upper limit thereof is preferably not more than 0.734, more preferably not more than 0.731, and even more preferably not more than 0.728. In addition, when the value of A1 is from 0.614 to 0.655, the value of A3 is preferably at least 0.735, more preferably at least 0.738, and even more preferably at least 0.741, and an upper limit thereof is preferably not more than 0.815, more preferably not more than 0.780, and even more preferably not more than 0.760. If the above value deviates from the above ranges, under all striking conditions, the ball may blow up, there may be a trajectory in which the ball does not carry, and an intended total distance may not be attainable.

[0143] The average value of the above A2 and A3, that is, the value of $(A2+A3)/2$ is at least 0.670, preferably at least 0.680, and more preferably at least 0.690, and an upper limit thereof is preferably not more than 0.783, more preferably not more than 0.775, and even more preferably not more than 0.765. If this value is too low, it becomes difficult for the ball to carry on shots with a driver (W #1) by amateur users, and the intended total distance may not be attainable. On the other hand, if the above value is too high, the ball trajectory may be blown up on shots with a driver (W #1) by amateur users, and the intended distance may not be attainable.

[0144] When the total volume of the dimples is denoted by D (mm.sup.3), and a deflection when the ball is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is denoted by B (mm), a value of D/B is preferably not more than 140, more preferably not more than 135, and even more preferably not more than 130. On the other hand, a lower limit thereof is preferably at least 100, more preferably at least 105, and even more preferably at least 110. This D/B means an index in which an appropriate distance suppression effect is produced on shots with a driver (W #1) by the longest hitters, and a good distance is easily obtained on shots by amateur users. If this value deviates from the above ranges, the intended distance may not be attainable on shots with a driver (W #1) by the longest hitters and by amateur users.

[0145] The golf ball of the present invention may be made to conform to the Rules of Golf for play. The inventive ball may be formed to a diameter which is such that the ball does not pass through a ring having an inner diameter of 42.672 mm and to a weight which is preferably between 45.0 and 45.93 g.

EXAMPLES

[0146] Hereinafter, the present invention is specifically described with reference to Examples and Comparative Examples, but the present invention is not limited to the following Examples.

Example 1 to 4 and Comparative Examples 1 to 13

[Formation of Core]

[0147] In Comparative Examples 6, 8, and 10 to 12, a rubber composition of each Example shown in Table 1 was prepared, and then vulcanization molding was performed under vulcanization conditions according to each Example shown in Table 1 to produce a solid core.

[0148] In Examples 1 to 4 and Comparative Examples 1 to 5, 7, 9, and 13, cores are produced based on formulations in Table 1 in the same manner as described above.

TABLE-US-00001 TABLE 1 Example Comparative Example Core formulation (pbw) 1 2 3 4 1 2 3

	4	5	Polybutadiene A	100	100	100	100	35	100	100	100	100	Polybutadiene B	Polybutadiene C	
Isoprene rubber	Styrene-butadiene	65	rubber	Zinc	acrylate	32.5	32.5	32.5	33.0	26.8	33.0	33.0	33.35		
26.0	Zinc methacrylate	1.0	Methacrylic acid	Zinc	stearate	2.0	2.0	2.0	2.0	2.0	2.0	5.0	Organic		
peroxide A	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.6	Organic peroxide B	0.6	Sulfur	0.025	0.025	0.025	
0.025	0.025	0.025	0.025	Water	0.3	0.3	0.3	0.2	0.4	0.2	0.2	0.3	Antioxidant	0.1	
0.1	0.1	0.1	0.1	Zinc oxide	28.3	28.3	28.3	28.1	26.4	28.1	28.1	11.8	14.2	Barium sulfate	
0.1	0.1	0.1	0.1	1.0	pentachlorothiophenol	Vulcanization	Temp.	150	150	150	150	150	150	150	
160	conditions (° C.)	Time	19	19	19	19	19	19	19	11	(min)	Comparative	Example	Core	
formulation	(pbw)	6	7	8	9	10	11	12	13	Polybutadiene A	100	100	100	100	
35	Polybutadiene B	20	20	Polybutadiene C	80	80	Isoprene rubber	5	Styrene-butadiene	65	rubber	Zinc	acrylate	37.0	
37.0	37.0	33.5	33.5	26.9	Zinc methacrylate	1.0	1.0	1.0	1.0	1.0	Methacrylic acid	23.5	Zinc	stearate	
2.0	2.0	Organic peroxide A	1.0	1.0	1.0	1.0	1.0	1.0	1.2	Organic peroxide B	Sulfur	0.025	0.025		
Water	0.4	0.4	0.4	0.4	0.6	0.6	0.4	Antioxidant	0.1	0.1	0.1	0.1	0.1	0.1	
0.2	Zinc oxide	14.8	14.8	14.8	19.3	19.3	16.5	23.5	Barium sulfate	1.0	Zinc salt of	1.0	1.0	1.0	
0.6	0.6	pentachlorothiophenol	Vulcanization	Temp.	150	150	150	150	160	160	160	163	conditions (° C.)		
Time	19	19	19	19	14	14	19	21	(min)						

[0165] Next, in Comparative Examples 6, 8, and 10 to 12, the intermediate layer was formed by injection molding a resin material No. 3 or No. 4 of the intermediate layer shown in Table 2 around the core surface using an injection mold. Subsequently, the cover was formed by injection molding the resin material No. 8 of the cover (outermost layer) shown in Table 2 around the intermediate layer-encased sphere using a separate injection mold. At this time, a predetermined large number of dimples described below were formed on the surface of the cover.

TABLE-US-00002 TABLE 2 Resin material (pbw) No. 1 No. 2 No. 3 No. 4 No. 5 No. 6 No. 7 No.

8 No. 9 HPF1000 56 100 Himilan 1605 44 50 50 50 Himilan 1601 37.5 Himilan 1557 15 37.5 Himilan 1706 15 35 50 AM7329 50 AM7318 85 AN4319 25 Titanium oxide 4 4 4 3 3 Barium sulfate 20 Magnesium stearate 1 Trimethylolpropane 1.1 1.1 TPU (1) 100 TPU (2) 100 [0167] Details of the blending components in Table 2 are as follows. [0168] “HPF 1000” ionomer resin manufactured by The Dow Chemical Company [0169] “Himilan 1605”, “Himilan 1601”, “Himilan 1557”, “Himilan 1706”, “AM7329”, and [0170] “AM7318” ionomer resins manufactured by Dow-Mitsui Polychemicals Co., Ltd. [0171] “AN4319” NUCREL™ manufactured by Dow-Mitsui Polychemicals Co., Ltd. [0172] Trade name “Precipitated Barium Sulfate 300” barium sulfate manufactured by Sakai Chemical Industry Co., Ltd. [0173] Trade name “Magnesium stearate G” magnesium stearate manufactured by NOF Corporation [0174] “Trimethylolpropane” (TMP) manufactured by Tokyo Chemical Industry Co., Ltd. [0175] Trade name “Pandex” ether-type thermoplastic polyurethane (TPU (1)), material hardness (Shore D) 50, manufactured by DIC Covestro Polymer Ltd.

[0176] Trade name “Pandex” ether-type thermoplastic polyurethane (TPU (2)), material hardness (Shore D) 47, manufactured by DIC Covestro Polymer Ltd.

[0177] For the dimples of the Examples and Comparative Examples, the following dimples (1) to (5) are used. Each dimple mode includes eight types of circular dimples of No. 1 to No. 8 having different diameters and depths. Details thereof are listed in Table 3 below. In addition, an arrangement mode (pattern) of the dimples (1) to (5) is illustrated in FIGS. 5A and 5B. FIG. 5A is a plan view of the dimples, and FIG. 5B is a side view thereof.

TABLE-US-00003	TABLE 3	Cylinder	Total volume	Diameter	Depth	Volume	volume ratio	SR	VR
of dimples	Type	Quantity	(mm)	(mm)	(mm.sup.3)	Vo (%)	(%)	(mm.sup.3)	Dimple (1)
No. 1	12	4.63	0.122	1.009	0.491	84	0.68	278	No. 2
No. 2	198	4.50	0.119	0.919	0.486	No. 3	36	3.92	0.115
No. 3	36	3.92	0.115	0.655	0.472	No. 4	12	2.87	0.086
No. 4	12	2.87	0.086	0.245	0.442	No. 5	36	4.49	0.126
No. 5	36	4.49	0.126	0.965	0.483	No. 6	24	3.92	0.124
No. 6	24	3.92	0.124	0.711	0.473	No. 7	6	3.31	0.133
No. 7	6	3.31	0.133	0.550	0.479	No. 8	6	3.21	0.132
No. 8	6	3.21	0.132	0.457	0.430	Total	330	Dimple (2)	No. 1
No. 1	12	4.69	0.144	1.220	0.493	86	0.81	331	No. 2
No. 2	198	4.54	0.140	1.100	0.486	No. 3	36	3.96	0.134
No. 3	36	3.96	0.134	0.766	0.467	No. 4	12	2.92	0.109
No. 4	12	2.92	0.109	0.322	0.441	No. 5	36	4.54	0.146
No. 5	36	4.54	0.146	1.141	0.485	No. 6	24	3.96	0.144
No. 6	24	3.96	0.144	0.827	0.468	No. 7	6	3.36	0.136
No. 7	6	3.36	0.136	0.590	0.491	No. 8	6	3.22	0.127
No. 8	6	3.22	0.127	0.433	0.421	Total	330	Dimple (3)	No. 1
No. 1	12	4.69	0.146	1.236	0.490	86	0.83	338	No. 2
No. 2	198	4.54	0.143	1.122	0.484	No. 3	36	3.96	0.137
No. 3	36	3.96	0.137	0.790	0.470	No. 4	12	2.92	0.106
No. 4	12	2.92	0.106	0.310	0.438	No. 5	36	4.54	0.149
No. 5	36	4.54	0.149	1.162	0.482	No. 6	24	3.96	0.147
No. 6	24	3.96	0.147	0.846	0.467	No. 7	6	3.38	0.138
No. 7	6	3.38	0.138	0.608	0.493	No. 8	6	3.28	0.132
No. 8	6	3.28	0.132	0.470	0.423	Total	330	Dimple (4)	No. 1
No. 1	12	4.70	0.149	1.252	0.487	86	0.85	345	No. 2
No. 2	198	4.55	0.146	1.144	0.483	No. 3	36	3.97	0.140
No. 3	36	3.97	0.140	0.815	0.472	No. 4	12	2.91	0.103
No. 4	12	2.91	0.103	0.298	0.435	No. 5	36	4.55	0.152
No. 5	36	4.55	0.152	1.183	0.480	No. 6	24	3.97	0.150
No. 6	24	3.97	0.150	0.864	0.467	No. 7	6	3.40	0.140
No. 7	6	3.40	0.140	0.627	0.495	No. 8	6	3.33	0.138
No. 8	6	3.33	0.138	0.510	0.426	Total	330	Dimple (5)	No. 1
No. 1	12	4.68	0.164	1.391	0.493	85	0.93	378	No. 2
No. 2	198	4.53	0.161	1.258	0.485	No. 3	36	3.95	0.154
No. 3	36	3.95	0.154	0.883	0.468	No. 4	12	2.90	0.114
No. 4	12	2.90	0.114	0.331	0.440	No. 5	36	4.53	0.168
No. 5	36	4.53	0.168	1.294	0.480	No. 6	24	3.95	0.165
No. 6	24	3.95	0.165	0.949	0.470	No. 7	6	3.36	0.154
No. 7	6	3.36	0.154	0.663	0.487	No. 8	6	3.26	0.152
No. 8	6	3.26	0.152	0.538	0.426	Total	330		

[Definition of Dimple]

[0178] Edge: highest point in cross section passing through center of a dimple [0179] Diameter: diameter of the flat plane circumscribed by the edge of a dimple [0180] Depth: maximum depth of a dimple from the flat plane circumscribed by the edge of the dimple [0181] SR: a ratio of a sum of the individual dimple surface areas, each defined by a flat plane circumscribed by an edge of a dimple, to a ball spherical surface area on the assumption that the ball has no dimples [0182] Dimple volume: a dimple volume under a flat plane circumscribed by an edge of a dimple [0183] Cylinder volume ratio: a ratio of the dimple volume to the cylinder volume having the same diameter as the dimple [0184] VR: a sum of the volumes of the individual dimples formed below the flat plane circumscribed by the edge of the dimple to the ball spherical volume on the assumption that the ball has no dimples

[0185] The ratio $CL1/CD1=A1$ of the lift coefficient $CL1$ at a Reynolds number of 218,000 and a spin rate of 2,800 rpm to the drag coefficient $CD1$, the ratio $CL2/CD2=A2$ of the lift coefficient

CL2 at a Reynolds number of 184,000 and a spin rate of 2,900 rpm to the drag coefficient CD2, and the ratio CL3/CD3=A3 of the lift coefficient CL3 at a Reynolds number of 158,000 and a spin rate of 3,100 rpm to the drag coefficient CD3 of the balls with the above dimples (1) to (5) formed on their cover surfaces are listed in the table below. These lift coefficients and drag coefficients are measured in accordance with the Indoor Test Range (ITR) defined by USGA.

TABLE-US-00004 TABLE 4 Dimple Dimple Dimple Dimple Dimple (1) (2) (3) (4) (5) CL1 0.151 0.147 0.146 0.145 0.143 CD1 0.230 0.237 0.238 0.239 0.244 CL1/CD1 = A1 0.657 0.620 0.613 0.607 0.586 CL2 0.168 0.162 0.161 0.160 0.156 CD2 0.233 0.240 0.241 0.242 0.246 CL2/CD2 = A2 0.721 0.675 0.668 0.661 0.634 CL3 0.190 0.182 0.181 0.179 0.173 CD3 0.242 0.245 0.246 0.247 0.250 CL3/CD3 = A3 0.785 0.743 0.734 0.725 0.692

[0186] For each resulting golf ball, various physical properties such as internal hardnesses at various positions of the core, outer diameters of the core and each layer-encased sphere, thicknesses and material hardnesses of each layer, surface hardnesses of each layer-encased sphere, and ball initial velocities are evaluated by the following methods, and are shown in Tables 5 to 8.

[Core Hardness Profile]

[0187] The core surface is spherical, but an indenter of a durometer is set substantially perpendicular to the spherical core surface, and a core surface hardness expressed on the Shore C scale is measured in accordance with ASTM D2240. With respect to the core center and a predetermined position of the core, the core is cut into hemispheres to obtain a flat cross-section, the hardness is measured by perpendicularly pressing the indenter of the durometer against a center portion and the predetermined positions shown in Tables 5 and 6, and the hardnesses at the center and each position are shown as Shore C hardness values. For the measurement of the hardness, a P2 Automatic Rubber Hardness Tester manufactured by Kobunshi Keiki Co., Ltd. equipped with a Shore C durometer is used. For the hardness value, a maximum value is read. All measurements are carried out in an environment of $23 \pm 2^\circ \text{C}$. It is noted that the numerical values in the table are Shore C hardness values.

[0188] In addition, in the core hardness profile, letting the Shore C hardness at the core center be C_c , the Shore C hardness at the midpoint M between the core center and the core surface be C_m , the respective Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm inward from the midpoint M be C_{m-2} , C_{m-4} , and C_{m-6} , the respective Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm outward from the midpoint M be C_{m+2} , C_{m+4} , and C_{m+6} , and the Shore C hardness at the core surface be C_s , the surface areas A to F are calculated as follows:

$$\frac{1}{2} \times 2 \times (C_{m-4} - C_{m-6}) \quad \text{surface area A:}$$

$$\frac{1}{2} \times 2 \times (C_{m-2} - C_{m-4}) \quad \text{surface area B:}$$

$$\frac{1}{2} \times 2 \times (C_m - C_{m-2}) \quad \text{surface area C:}$$

$$\frac{1}{2} \times 2 \times (C_{m+2} - C_m) \quad \text{surface area D:}$$

$$\frac{1}{2} \times 2 \times (C_{m+4} - C_{m+2}) \quad \text{surface area E:}$$

$$\frac{1}{2} \times 2 \times (C_{m+6} - C_{m+4}) \quad \text{surface area F: [0189] and the values of the following seven expressions are determined.}$$

$$\text{Surface area A} + \text{surface area B} \quad (1)$$

$$\text{Surface area B} + \text{surface area C} \quad (2)$$

Surface area D +surface area E (3)

(Surface area D +surface area E)−(surface area A +surface area B) (4)

(Surface area D +surface area E)−(surface area B +surface area C) (5)

{(Surface area D +surface area E)−(surface area A +surface area B)}×(C_s − C_c) (6)

{(Surface area D +surface area E)−(surface area B +surface area C)}×(C_s − C_c) (7)

[0190] The surface areas A to F in the core hardness profile are described in FIG. 2, which shows a graph that illustrates surface areas A to F using the core hardness profile data from Example 1.

[0191] In addition, FIGS. 3 and 4 show graphs of core hardness profiles for Examples 1 to 4 and Comparative Examples 1 to 13.

[Diameters of Core and of Intermediate Layer-Encased Sphere]

[0192] At a temperature adjusted to $23.9\pm1^\circ\text{C}$. for at least three hours or more in a thermostatic bath, five random places on the surface are measured in a room with a temperature of $23.9\pm2^\circ\text{C}$., and, using an average value of these measurements as a measured value of each sphere, an average value for the diameter of 10 such spheres is determined.

[Ball Diameter]

[0193] At a temperature adjusted to $23.9\pm1^\circ\text{C}$. for at least three hours or more in a thermostatic bath, a diameter at 15 random dimple-free places is measured in a room at a temperature of $23.9\pm2^\circ\text{C}$., and, using an average value of these measurements as a measured value of one ball, an average value for the diameter of 10 balls is determined.

[Deflections of Core and Ball]

[0194] Each subject layer-encased sphere of the core or the ball is placed on a hard plate, and the deflection when compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is measured. It is noted that the deflection in each case is a measurement value measured in a room at a temperature of $23.9\pm2^\circ\text{C}$. after temperature adjustment to $23.9\pm1^\circ\text{C}$. for at least three hours or more in a thermostatic bath. As a measuring device, a high-load compression tester manufactured by MU Instruments Trading Corp. is used, and a down speed of a pressure head that compresses the core or the ball is set to 10 mm/s.

[Material Hardnesses of Intermediate Layer and Cover (Shore C and Shore D Hardnesses)]

[0195] The resin material of each layer is molded into a sheet having a thickness of 2 mm and left at a temperature of $23\pm2^\circ\text{C}$. for two weeks. At the time of measurement, three such sheets are stacked together. The Shore C hardness and the Shore D hardness are each measured with a Shore C durometer and a Shore D durometer conforming to the ASTM D2240 standard. For the measurement of the hardness, the P2 Automatic Rubber Hardness Tester manufactured by Kobunshi Keiki Co., Ltd. to which a Shore C durometer or a Shore D durometer is mounted is used. For the hardness value, a maximum value is read. The measurement method is in accordance with the ASTM D2240 standard.

[Surface Hardnesses of Intermediate Layer-Encased Sphere and of Ball]

[0196] A measurement is performed by perpendicularly pressing the indenter against the surface of each sphere. It is noted that a surface hardness of a ball (cover) is a measured value at a dimple-free area (land) on the surface of the ball. The Shore C hardness and the Shore D hardness are each measured with a Shore C durometer and a Shore D durometer conforming to the ASTM D2240 standard. For the measurement of the hardness, the P2 Automatic Rubber Hardness Tester manufactured by Kobunshi Keiki Co., Ltd. to which a Shore C durometer or a Shore D durometer is mounted is used. For the hardness value, a maximum value is read. The measurement method is in accordance with the ASTM D2240 standard.

[Initial Velocity of Ball]

[0197] The initial velocity of each sphere is measured at a temperature of $23.9 \pm 2^\circ \text{C}$. using a device for measuring COR manufactured by Hye Precision Products of the same type as the R&A. The measurement principle is as follows.

[0198] An air pressure is changed to four stages of 35.5 psi, 36.5 psi, 39.5 psi, and 40.5 psi, and a ball is fired at four stages of incident velocity by respective air pressures, collided with a barrier, and its COR is measured. That is, a correlation equation between the incident velocity and the COR is created by changing the air pressure in four stages. Similarly, a correlation equation between the incident velocity and a contact time is created.

[0199] Then, from these correlation equations, the COR (coefficient of restitution) and the contact time (μs) at an incident velocity of 43.83 m/s are determined and substituted into the following initial velocity conversion equation to calculate an initial velocity of each sphere.

$$[00009] IV = 136.8 + 136.3e + 0.019tc$$

[0200] [Here, e is a coefficient of restitution, and the is a contact time (μs) at a collision speed of 143.8 ft/s (43.83 m/s).]

[0201] In the initial velocity measurement of the balls of all examples, a barrel diameter of 43.18 mm is selected.

TABLE-US-00005 TABLE 5 Example Comparative Example														1	2	3	4	1	2	3	4	Ball structure (piece)												
3P	3P	3P	3P	3P	3P	3P	3P	2P	Core	Outer diameter (mm)	37.30	37.30	37.30	37.30	37.30	37.30	37.30	37.30																
39.30	Weight (g)	32.8	32.8	32.8	32.8	32.8	32.8	32.8	35.8	Deflection (mm)	3.40	3.40	3.40	3.30	3.40	3.40	3.40	3.40																
3.40	3.40	3.12	Cs [surface] (Shore C)	85.5	85.5	85.5	86.5	77.0	85.5	85.5	88.3	Cm + 6 (Shore C)	75.3	75.3	75.3	76.1	76.3	75.3	75.3	77.5	Cm + 4 (Shore C)	69.5	69.5	69.5	70.1	75.2	69.5	69.5	71.1					
63.2	64.0	69.7	63.2	63.2	65.4	Cm - 2 (Shore C)	63.0	63.0	63.0	63.8	65.7	63.0	63.0	65.2	Cm - 4 (Shore C)	62.2	62.2	62.2	63.1	63.4	62.2	62.2	64.7	Cm - 6 (Shore C)	61.8	61.8	61.8	62.8	61.7	61.8				
61.8	64.6	Cc [center] (Shore C)	61.5	61.5	61.5	62.6	60.1	61.5	61.5	64.6	Cs - Cc (Shore C)	24.0	24.0	23.9	16.9	24.0	24.0	23.7	(Cs - Cc)/(Cm - Cc)	14.1	14.1	14.1	17.1	1.8	14.1	14.1	29.6							
5.0	5.0	5.0	5.0	2.3	5.0	5.0	5.0	Surface area F	5.8	5.8	5.8	6.0	1.1	5.8	5.8	6.4	Surface area A + surface area B	1.2	1.2	1.2	1.0	4.0	1.2	1.2	0.6	Surface area B + surface area C	1.0	1.0	1.0	0.9	6.3	1.0	1.0	0.7
5.3	5.2	-0.8	5.3	5.3	5.0	{(Surface areas: D + E) - (surface areas: A + 122	122	122	122	25	122	122	121 B)} × (Cs - Cc)	{(Surface areas: D + E) - (surface areas: B + 127	127	127	124	-14	127	127	119 C)} × (Cs - Cc)													

TABLE-US-00006 TABLE 6 Comparative Example														5	6	7	8	9	10	11	12	13	Ball structure (piece)	2P								
3P	3P	3P	3P	3P	3P	3P	3P	2P	Core	Outer diameter (mm)	39.70	38.65	38.65	38.65	38.65	38.06	38.06	38.64	39.80	Weight (g)	37.7	35.1	35.1	35.1	35.1	33.8	33.8	35.1	37.0	Deflection (mm)	4.50	2.92
2.92	2.92	2.92	2.92	4.13	4.13	2.93	2.63	Cs [surface] (Shore C)	76.0	87.4	87.4	87.4	87.4	87.4	86.3	86.3	81.5	84.3	Cm + 6 (Shore C)	66.5	80.4	80.4	80.4	80.4	74.1	74.1	80.5	79.0	Cm + 4 (Shore C)	64.9	75.8	
75.8	75.8	75.8	65.9	65.9	79.0	75.9	Cm + 2 (Shore C)	63.4	70.8	70.8	70.8	70.8	70.8	61.3	61.3	76.2	72.8	Cm [intermediate] (Shore C)	62.9	66.3	66.3	66.3	66.3	61.0	61.0	72.3	70.8	Cm - 2 (Shore C)	62.4	65.6		
65.6	65.6	65.6	65.6	61.4	61.4	68.4	69.8	Cm - 4 (Shore C)	61.3	64.9	64.9	64.9	64.9	61.0	61.0	66.9	68.4	Cm - 6 (Shore C)	59.7	63.3	63.3	63.3	63.3	60.1	60.1	65.4	66.3	Cc [center] (Shore C)	56.2	62.6		
62.6	62.6	62.6	62.6	57.9	57.9	62.7	60.7	Cs - Cc (Shore C)	19.8	24.8	24.8	24.8	24.8	24.8	28.4	28.4	18.8	23.6	(Cs - Cc)/(Cm - Cc)	3.0	6.7	6.7	6.7	6.7	9.2	9.2	2.0	2.3	Surface area A	1.6	1.6	
1.6	1.6	1.6	1.6	1.6	0.9	0.9	1.5	2.1	Surface area B	1.1	0.7	0.7	0.7	0.7	0.4	0.4	1.5	1.4	Surface area C	0.5	0.7	0.7	0.7	0.7	-0.4	-0.4	3.9	1.0	Surface area D	0.5	4.5	
4.5	4.5	4.5	4.5	4.5	0.3	0.3	3.9	2.0	Surface area E	1.5	5.0	5.0	5.0	5.0	4.6	4.6	2.8	3.1	Surface area F	1.6	4.6	4.6	4.6	4.6	8.2	8.2	1.5	3.1	Surface area A + surface area B	2.7	2.3	
2.3	2.3	2.3	1.3	1.3	3.0	3.5	Surface area B + surface area C	1.6	1.4	1.4	1.4	1.4	1.4	0.0	0.0	5.4	2.4															

Surface area D + surface area E 2.0 9.5 9.5 9.5 9.5 4.9 4.9 6.7 5.1 (Surface areas: D + E) – (surface areas: A + B) –0.7 7.2 7.2 7.2 7.2 3.6 3.6 3.7 1.6 (Surface areas: D + E) – (surface areas: B + C) 0.4 8.1 8.1 8.1 8.1 4.9 4.9 1.3 2.7 {(Surface areas: D + E) – (surface areas: A + –14 179 179 179 179 102 102 70 38 B)} × (Cs – Cc) {(Surface areas: D + E) – (surface areas: B + 8 201 201 201 201 139 139 24 64 C)} × (Cs – Cc)

TABLE-US-00007 TABLE 7 Example Comparative Example 1 2 3 4 1 2 3 4 Intermediate Material No. 1 No. 1 No. 1 No. 2 No. 1 No. 1 No. 1 — layer Thickness (mm) 1.35 1.35 1.35 1.35 1.35 1.35 1.35 — Material hardness (Shore C) 90 90 90 84 90 90 90 — Material hardness (Shore D) 59 59 59 54 59 59 59 — Intermediate Outer diameter (mm) 40.00 40.00 40.00 40.00 40.00 40.00 40.00 — layer-encased Weight (g) 38.9 38.9 38.9 38.9 38.9 38.9 38.9 — sphere Surface hardness (Shore C) 94 94 94 90 94 94 94 — Surface hardness (Shore D) 63 63 63 59 63 63 63 — Cover Material No. 5 No. 5 No. 5 No. 5 No. 5 No. 5 No. 6 Thickness (mm) 1.35 1.35 1.35 1.35 1.35 1.35 1.35 1.70 Material hardness (Shore C) 93 93 93 93 93 93 93 94 Material hardness (Shore D) 64 64 64 64 64 64 64 65 Dimple Type (2) (3) (4) (2) (1) (1) (5) (4) Quantity 330 330 330 330 330 330 330 330 Surface area coverage ratio: SR (%) 86 86 86 86 84 84 85 86 Volume occupancy ratio: VR (%) 0.81 0.83 0.85 0.81 0.68 0.68 0.93 0.85 Total volume of dimples (mm.sup.3) 331 338 345 331 278 278 378 345 A1: CL1/CD1 0.620 0.613 0.607 0.620 0.657 0.657 0.586 0.607 A2: CL2/CD2 0.675 0.668 0.661 0.675 0.721 0.721 0.634 0.661 A3: CL3/CD3 0.743 0.734 0.725 0.743 0.785 0.785 0.692 0.725 Average value of A2 and A3 0.709 0.701 0.693 0.709 0.753 0.753 0.663 0.693 Ball Outer diameter (mm) 42.70 42.70 42.70 42.70 42.70 42.70 42.70 42.70 Weight (g) 45.50 45.50 45.50 45.50 45.50 45.50 45.50 Deflection (mm) 2.70 2.70 2.70 2.70 2.70 2.70 2.70 2.71 Initial velocity (m/s) 77.2 77.2 77.2 77.2 74.0 77.2 77.2 77.0 Initial velocity/deflection 28.6 28.6 28.6 27.4 28.6 28.6 28.4 Surface hardness (Shore C) 98 98 98 98 98 98 98 98 Surface hardness (Shore D) 70 70 70 70 70 70 70 71 Total volume of dimples/deflection of ball (mm.sup.2) 123 125 128 123 103 103 140 127 Ball surface hardness - surface hardness of 4 4 4 8 4 4 4 — intermediate layer-encased sphere (Shore C) Surface hardness of intermediate layer-encased 9 9 9 4 17 9 9 — sphere - core surface hardness (Shore C) Surface hardness of intermediate layer-encased 33 33 33 27 34 33 33 — sphere - core center hardness (Shore C) Ball surface hardness - 13 13 13 12 21 13 13 10 core surface hardness (Shore C) Ball surface hardness - 37 37 37 35 38 37 37 33 core center hardness (Shore C) Deflection of core - deflection of ball (mm) 0.70 0.70 0.70 0.60 0.70 0.70 0.70 0.41 Deflection of core/deflection of ball 1.26 1.26 1.26 1.22 1.26 1.26 1.26 1.15 Core diameter/ball diameter 0.874 0.874 0.874 0.874 0.874 0.874 0.874 0.920

TABLE-US-00008 TABLE 8 Comparative Example 5 6 7 8 9 10 11 12 13 Intermediate Material — No. 3 No. 3 No. 3 No. 3 No. 4 No. 4 No. 3 — layer Thickness (mm) — 1.17 1.17 1.17 1.17 1.47 1.47 1.21 — Material hardness (Shore C) — 94 94 94 94 94 94 94 — Material hardness (Shore D) — 67 67 67 67 65 65 67 — Intermediate Outer diameter (mm) — 40.99 40.99 40.99 40.99 41.00 41.00 41.06 — layer-encased Weight (g) — 40.6 40.6 40.6 40.6 40.7 40.7 40.8 — sphere Surface hardness (Shore C) — 97 97 97 97 97 97 97 — Surface hardness (Shore D) — 71 71 71 71 71 71 71 — Cover Material No. 7 No. 8 No. 8 No. 8 No. 8 No. 8 No. 8 No. 8 No. 9 Thickness (mm) 1.50 0.85 0.85 0.85 0.85 0.84 0.84 0.81 1.46 Material hardness (Shore C) 84 71 71 71 71 71 71 71 67 Material hardness (Shore D) 56 50 50 50 50 50 50 50 47 Dimple Type (4) (4) (2) (1) (5) (1) (4) (1) (2) Quantity 330 330 330 330 330 330 330 330 330 Surface area coverage ratio: SR (%) 86 86 86 84 85 84 86 84 86 Volume occupancy ratio: VR (%) 0.85 0.85 0.81 0.68 0.93 0.68 0.85 0.68 0.81 Total volume of dimples (mm.sup.3) 345 345 331 278 378 278 345 278 331 A1: CL1/CD1 0.607 0.607 0.620 0.657 0.586 0.657 0.607 0.657 0.620 A2: CL2/CD2 0.661 0.661 0.675 0.721 0.634 0.721 0.661 0.721 0.675 A3: CL3/CD3 0.725 0.725 0.743 0.785 0.692 0.785 0.725 0.785 0.743 Average value of A2 and A3 0.693 0.693 0.709 0.753 0.663 0.753 0.693 0.753 0.709 Ball Outer diameter (mm) 42.70 42.69 42.69 42.69 42.69 42.68 42.69 42.68 42.72 Weight (g) 45.40 45.49 45.52 45.55 45.52 45.60 45.52 45.51 45.66 Deflection (mm) 4.00 2.37 2.35 2.32 2.35 2.96 2.98 2.38 2.51 Initial velocity (m/s) 77.0 77.0 77.1 77.2 77.1 76.9 77.0 73.1 73.4 Initial

velocity/deflection 19.3 32.5 32.6 33.3 32.8 26.0 25.8 30.7 29.2 Surface hardness (Shore C) 92 87 87 87 86 86 87 79 Surface hardness (Shore D) 62 61 61 61 61 60 60 61 53 Total volume of dimples/deflection of ball (mm.sup.2) 86 145 141 120 161 94 116 117 132 Ball surface hardness - surface hardness of — -10 -10 -10 -10 -11 -11 -10 — intermediate layer-encased sphere (Shore C) Surface hardness of intermediate layer-encased — 10 10 10 10 11 11 16 — sphere - core surface hardness (Shore C) Surface hardness of intermediate layer-encased — 34 34 34 34 39 39 34 — sphere - core center hardness (Shore C) Ball surface hardness - 16 0 0 0 0 0 0 6 -5 core surface hardness (Shore C) Ball surface hardness - 36 24 24 24 24 28 28 24 18 core center hardness (Shore C) Deflection of core - deflection of ball (mm) 0.50 0.55 0.57 0.60 0.57 1.17 1.15 0.55 0.12 Deflection of core/deflection of ball 1.13 1.23 1.24 1.26 1.24 1.40 1.39 1.23 1.05 Core diameter/ball diameter 0.930 0.905 0.905 0.905 0.905 0.892 0.892 0.905 0.932

[0202] The flight (W #1 and I #6) and the controllability on approach shots of each golf ball are evaluated by the following methods. The results are shown in Table 9.

[Evaluation of Flight (W #1, HS 54 m/s)]

[0203] A driver is mounted on a golf swing robot, and a spin rate and a distance traveled (total) by a ball when struck at a head speed (HS) of 54 m/s are measured. The club used is a TOUR B XD-5 Driver/loft angle 8.5° (2017 model) manufactured by Bridgestone Sports Co., Ltd. and is evaluated according to the following rating criteria.

[Rating Criteria]

[0204] Good: Total compared to Comparative Example 8 is not more than-8.0 m, and at least-14.0 m. [0205] Fair: Total compared with Comparative Example 8 is less than-14.0 m. [0206] NG: Total compared with Comparative Example 8 is more than-8.0 m.

[Evaluation of Flight (W #1, HS 40 m/s)]

[0207] The driver is mounted on the golf swing robot, and the spin rate and the distance traveled (total) by a ball when struck at a head speed (HS) of 40 m/s are measured. The club used is a J015 Driver/loft angle 9.5° (2016 model) manufactured by Bridgestone Sports Co., Ltd. and is evaluated according to the following rating criteria.

[Rating Criteria]

[0208] Good: Total compared with Comparative Example 8 is at least 2.0 m. [0209] Fair: Total compared with Comparative Example 8 is at least-3.0 m and less than 2.0 m. [0210] NG: Total compared with Comparative Example 8 is less than-3.0 m.

[Evaluation of Flight (I #6, HS 42 m/s)]

[0211] When a number six iron (I #6) is mounted on the golf swing robot and a ball is struck at an HS of 42 m/s, a spin rate and a distance traveled (total) are measured. The club used is a JGR Forged I #6 (2016 model) manufactured by Bridgestone Sports Co., Ltd. and is evaluated according to the following rating criteria.

[Rating Criteria]

[0212] Good: Total compared with Comparative Example 8 is at least 5.0 m. [0213] Fair: Total compared with Comparative Example 8 is at least 0 m and less than 5.0 m. [0214] NG: Total compared with Comparative Example 8 is less than 0 m.

[Evaluation of Flight (I #6, HS 35 m/s)]

[0215] When a number six iron (I #6) is mounted on the golf swing robot and a ball is struck at an HS of 35 m/s, a spin rate and a distance traveled (total) are measured. The club used is a JGR Forged I #6 (2016 model) manufactured by Bridgestone Sports Co., Ltd. and is evaluated according to the following rating criteria.

[Rating Criteria/Total]

[0216] Good: Total compared with Comparative Example 8 is at least 3.0 m. [0217] Fair: Total compared with Comparative Example 8 is at least -3.0 m and less than 3.0 m. [0218] NG: Total compared with Comparative Example 8 is less than -3.0 m.

[Evaluation of Launch Angle on Approach Shots]

[0219] A judgment is made based on a launch angle when a sand wedge is mounted on the golf swing robot and a ball is struck at a head speed (HS) of 15 m/s. The launch angle immediately after the ball is struck is measured by a device for measuring initial conditions. The sand wedge used is a TOURSTAGE TW-03 (loft angle 57°) 2002 model manufactured by Bridgestone Sports Co., Ltd.

[Rating Criteria]

[0220] Good: Launch angle is at least 35.0°. [0221] NG: Launch angle is less than 35.0°.

TABLE-US-00009 TABLE 9 Example Comparative Example 1 2 3 4 1 2 3 4 5 Flight W#1 Spin rate (rpm) 2,395 2,395 2,395 2,308 2,545 2,395 2,395 2,423 2,479 HS Total (m) 270.6 269.7 268.7 269.1 271.5 278.8 256.8 268.6 267.0 54 m/s Total (m) compared -10.9 -11.8 -12.8 -12.4 -10.0 -2.7 -24.7 -12.9 -14.5 with Comp. Ex. 8 Rating Good Good Good Good Good NG Fair Good Fair W#1 Spin rate (rpm) 2,740 2,740 2,740 2,662 2,840 2,740 2,740 2,754 2,692 HS Total (m) 201.0 201.5 201.9 201.1 189.8 199.8 200.1 200.0 198.1 40 m/s Total (m) compared 2.7 3.2 3.6 2.8 -8.5 1.5 1.8 1.7 -0.2 with Comp. Ex. 8 Rating Good Good Good Good NG Fair Fair Fair Fair I#6 Spin rate (rpm) 5,229 5,229 5,229 5,430 5,629 5,229 5,229 5,271 5,143 HS Total (m) 185.7 185.4 185.0 185.6 170.1 182.5 181.6 183.2 184.3 42 m/s Total (m) compared 7.6 7.3 6.9 7.5 -8.0 4.4 3.5 5.1 6.2 with Comp. Ex. 8 Rating Good Good Good Good NG Fair Fair Good Good I#6 Spin rate (rpm) 5,099 5,099 5,099 5,294 5,399 5,099 5,099 5,139 4,933 HS Total (m) 143.2 144.3 145.3 143.1 134.8 143.5 145.7 143.9 144.8 35 m/s Total (m) compared 3.1 4.2 5.2 3.0 -5.3 3.4 5.6 3.8 4.7 with Comp. Ex. 8 Rating Good Good Good Good NG Good Good Good Good Approach SW Launch angle (°) 41.2 41.2 41.2 41.2 40.2 41.2 41.2 41.1 36.7 HS Rating Good Good Good Good Good Good Good Good Good 15 m/s Comparative Example 6 7 8 9 10 11 12 13 Flight W#1 Spin rate (rpm) 2,801 2,794 2,786 2,794 2,542 2,545 2,981 3,149 HS Total (m) 271.5 273.3 281.5 259.6 276.1 268.8 269.3 259.8 54 m/s Total (m) compared -10.0 -8.2 0.0 -21.9 -5.4 -12.7 -12.2 -21.7 with Comp. Ex. 8 Rating Good Good NG Fair NG Good Good Fair W#1 Spin rate (rpm) 3,177 3,166 3,154 3,166 2,912 2,900 3,372 3,629 HS Total (m) 200.4 199.5 198.3 198.6 199.5 200.4 184.9 179.0 40 m/s Total (m) compared 2.1 1.2 0.0 0.3 1.2 2.1 -13.4 -19.3 with Comp. Ex. 8 Rating Good Fair Fair Fair Fair Good NG NG I#6 Spin rate (rpm) 5,780 5,751 5,721 5,751 4,948 5,039 6,249 6,354 HS Total (m) 180.6 181.3 178.1 177.2 184.4 186.6 163.4 167.1 42 m/s Total (m) compared 2.5 3.2 0.0 -0.9 6.3 8.5 -14.7 -11.0 with Comp. Ex. 8 Rating Fair Fair Fair NG Good Good NG NG I#6 Spin rate (rpm) 5,448 5,468 5,487 5,468 4,724 4,807 5,881 6,067 HS Total (m) 141.9 139.8 140.1 142.3 144.7 146.0 129.8 131.1 35 m/s Total (m) compared 1.8 -0.3 0.0 2.2 4.6 5.9 -10.3 -9.0 with Comp. Ex. 8 Rating Fair Fair Fair Fair Good Good NG NG Approach SW Launch angle (°) 32.5 32.5 32.6 32.5 32.8 32.8 31.5 32.6 HS Rating NG NG NG NG NG NG NG NG NG 15 m/s

[0222] As shown in the results in Table 9, the golf balls of Comparative Examples 1 to 13 are inferior in the following respects to the golf balls according to the present invention (Examples).

[0223] In Comparative Example 1, the dimple volume occupancy ratio VR is smaller than 0.75%, and A1 is larger than 0.655. In addition, the initial velocity of the ball is lower than 76.0 m/s. As a result, the distance on shots with a driver (W #1) at a head speed (HS) of 40 m/s is inferior, and the distance on shots with a number six iron (I #6) is inferior.

[0224] In Comparative Example 2, the dimple volume occupancy ratio VR is smaller than 0.75%, and A1 is larger than 0.655. As a result, the distance on shots with a driver (W #1) at a head speed (HS) of 54 m/s is longer, and does not conform to the new ODS rules.

[0225] In Comparative Example 3, the dimple volume occupancy ratio VR is larger than 0.89%. In addition, A1 is smaller than 0.590, and (A2+A3)/2 is smaller than 0.670. As a result, the distance is inferior under striking conditions with a driver (W #1) at head speeds (HS) of both 54 m/s and 40 m/s.

[0226] Comparative Example 4 is a golf ball having a two-piece structure without an intermediate layer. As a result, the distance is inferior under striking conditions with a driver (W #1) at a head speed (HS) of 40 m/s.

[0227] Comparative Example 5 is a golf ball having a two-piece structure without an intermediate layer, and a value obtained by dividing a total volume of dimples by a deflection of the ball at a load of 10 to 130 kgf is smaller than 100. As a result, a distance on shots with a driver (W #1) at a head speed (HS) of 54 m/s is excessively reduced, and a distance on shots with a driver (W #1) at an HS of 40 m/s is also inferior.

[0228] In Comparative Example 6, the value obtained by dividing the total volume of the dimples by the deflection of the ball under a load of 10 to 130 kgf is larger than 140, and the ball surface hardness is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, a distance on shots with an iron (I #6) is inferior, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0229] In Comparative Example 7, the value obtained by dividing the total volume of the dimples by the deflection of the ball under a load of 10 to 130 kgf is larger than 140, and the ball surface hardness is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, a distance on shots with an iron (I #6) is inferior, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0230] Comparative Example 8 is one of embodiments currently used by male professionals. The surface hardness of the ball is smaller than the surface hardness of the intermediate layer-encased sphere. The dimple volume occupancy ratio VR is smaller than 0.75%, and A1 is larger than 0.655. As a result, the distance on shots with a driver (W #1) at a head speed (HS) of 54 m/s is longer, and does not conform to the new ODS rules. Furthermore, distances on shots with a driver (W #1) at an HS of 40 m/s and with an iron (I #6) are inferior, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0231] In Comparative Example 9, the dimple volume occupancy ratio VR is larger than 0.89%. In addition, A1 is smaller than 0.590, and $(A2+A3)/2$ is smaller than 0.670. Furthermore, the value obtained by dividing the total volume of the dimples by the deflection of the ball under a load of 10 to 130 kgf is larger than 140, and the ball surface hardness is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, a distance on shots with a driver at a head speed (HS) of 54 m/s is excessively reduced, and distances on shots with a driver at an HS of 40 m/s with an iron (I #6) at an HS of 42 m/s are also inferior. In addition, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0232] In Comparative Example 10, the dimple volume occupancy ratio VR is smaller than 0.75%, and A1 is larger than 0.655. Furthermore, the value obtained by dividing the total volume of the dimples by the deflection of the ball under a load of 10 to 130 kgf is smaller than 100, and the ball surface hardness is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, the distance on shots with a driver (W #1) at a head speed (HS) of 54 m/s is longer, and does not conform to the new ODS rules. As a result, distances on shots with a driver at an HS of 40 m/s and with an iron (I #6) are inferior, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0233] In Comparative Example 11, the surface hardness of the ball is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, the launch angle on approach shots was too low, and the ball was difficult for an amateur.

[0234] In Comparative Example 12, the dimple volume occupancy ratio VR is smaller than 0.75%, and A1 is larger than 0.655. Furthermore, the initial velocity of the ball is smaller than 76.0 m/s, and the ball surface hardness is smaller than the surface hardness of the intermediate layer-encased sphere. As a result, distances on shots with a driver (W #1) at an HS of 40 m/s and with an iron (I #6) are inferior, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0235] Comparative Example 13 corresponds to a golf ball having a two-piece structure for a driving range, the initial velocity of the ball is less than 76.0 m/s, and the cover material is made of urethane. As a result, the actual initial velocity decreases under all striking conditions. A distance

on shots with a driver (W #1) at a head speed (HS) of 54 m/s becomes too short, and the distance also decreases under other striking conditions. In addition, the launch angle on approach shots is too low, and the ball is difficult for an amateur.

[0236] Japanese Patent Application No. 2024-033917 is incorporated herein by reference. Although some preferred embodiments have been described, many modifications and variations may be made thereto in light of the above teachings. It is therefore to be understood that the invention may be practiced otherwise than as specifically described without departing from the scope of the appended claims.

Claims

1. A multi-piece solid golf ball comprising a core, an intermediate layer, and a cover, wherein a large number of dimples are formed on an outside surface of the cover, a relationship between a surface hardness of an intermediate layer-encased sphere and a surface hardness of the ball satisfies the following condition:

Surface hardness of ball > surface hardness of intermediate layer-encased sphere where hardness means Shore C hardness, an initial velocity of the ball is from 76.0 to 77.724 m/s, and where a ratio CL1/CD1 of a lift coefficient CL1 at a Reynolds number of 218,000 and a spin rate of 2,800 rpm to a drag coefficient CD1 is denoted by A1, a ratio CL2/CD2 of a lift coefficient CL2 at a Reynolds number of 184,000 and a spin rate of 2,900 rpm to a drag coefficient CD2 is denoted by A2, and a ratio CL3/CD3 of a lift coefficient CL3 at a Reynolds number of 158,000 and a spin rate of 3,100 rpm to a drag coefficient CD3 is denoted by A3, the following two conditions are satisfied:

$0.59 \leq A1 \leq 0.655$, and $(A2 + A3) / 2 \geq 0.67$, and a volume occupancy ratio VR of the dimples is from 0.75 to 0.89%, and the following condition is satisfied: $100 \leq D / B \leq 140$ where a total volume of the dimples is denoted by D (mm.sup.3) and a deflection when the ball is compressed under a final load of 1,275 N (130 kgf) from an initial load of 98 N (10 kgf) is denoted by B (mm).

2. The multi-piece solid golf ball of claim 1, wherein the value of A1 is from 0.590 to 0.613, the value of A2 is from 0.635 to 0.668, and the value of A3 is from 0.695 to 0.734.

3. The multi-piece solid golf ball of claim 1, wherein the value of A1 is from 0.614 to 0.655, the value of A2 is from 0.669 to 0.750, and the value of A3 is from 0.735 to 0.815.

4. The multi-piece solid golf ball of claim 1, wherein the value of $(A2+A3)/2$ is from 0.670 to 0.783.

5. The multi-piece solid golf ball of claim 1, wherein the cover is formed of an ionomer resin as a chief material.

6. The multi-piece solid golf ball of claim 1, wherein the core has a hardness profile in which, letting the Shore C hardness at a core center be Cc, the Shore C hardness at a midpoint M between the core center and a core surface be Cm, the Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm inward from the midpoint M be Cm-2, Cm-4, and Cm-6 respectively, the Shore C hardnesses at positions 2 mm, 4 mm, and 6 mm outward from the midpoint M be Cm+2, Cm+4, and Cm+6 respectively, and the Shore C hardness at the core surface be Cs, and defining surface areas A to F as follows:

$\frac{1}{2} \times 2 \times (Cm-4 - Cm-6)$ surface area A:

$\frac{1}{2} \times 2 \times (Cm-2 - Cm-4)$ surface area B:

$\frac{1}{2} \times 2 \times (Cm - Cm-2)$ surface area C:

$\frac{1}{2} \times 2 \times (Cm+2 - Cm)$ surface area D:

$\frac{1}{2} \times 2 \times (Cm+4 - Cm+2)$ surface area E:

$\frac{1}{2} \times 2 \times (Cm+6 - Cm+4)$ surface area F: the following condition is satisfied:

$\{(\text{surface area } D + \text{surface area } E) - (\text{surface area } A + \text{surface area } B)\} \geq 4.0$.

7. The multi-piece solid golf ball of claim 6, wherein the core has a hardness profile in which the

following condition is satisfied: $(C_s - C_c) \geq 20$.

8. The multi-piece solid golf ball of claim 6, wherein the core has a hardness profile in which the following condition is satisfied: $(C_s - C_c) / (C_m - C_c) \geq 3.0$.

9. The multi-piece solid golf ball of claim 6, wherein in the core hardness profile, an upper limit of the formula $\{(\text{surface area D} + \text{surface area E}) - (\text{surface area A} + \text{surface area B})\}$ is 15.0.

10. The multi-piece solid golf ball of claim 1, wherein the core is formed of a rubber composition containing the following components (A) to (D): (A) a base rubber, (B) an organic peroxide, (C) water or a monocarboxylic acid metal salt, and (D) sulfur.
